

**LAUNDRYLOGIX: STREAMLINING SERVICE TRANSACTIONS THROUGH A
DIGITAL POINT OF SALE SYSTEM FOR ELYSSA LAUNDRY CENTER**

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of the Requirements for the Degree
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ABSTRACT

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7.1 Rationale/Background of the Study

The study addressed operational challenges at Elyssa Laundry Center, which relied on manual, paper-based processes for transactions, machine assignment, inventory monitoring, and customer communication that caused misplaced records, inconsistent pricing, limited performance visibility, and no real-time status updates. Drawing on literature on digital POS systems, web-based management, data and visual analytics, and user-centered design, the researchers proposed LaundryLogix, a web-based digital POS with a customer mobile application to streamline transactions, enhance monitoring, and provide clearer communication through in-app notifications.

7.2 Summary

The study aimed to design and develop LaundryLogix, a digital POS system for Elyssa Laundry Center, by



defining its information requirements, architectural framework, key features, and usability.

1. to identify the information requirements of the system;
2. to determine to architectural framework of the system;
3. to identify the features of the system; and
4. to measure the extent of usability of the system.

The project used Scrum as the main development approach, organizing work into short sprints driven by a prioritized product backlog of user stories. This iterative setup enabled frequent feedback from the owner and customers, allowing continuous refinement of both the POS and mobile application.

7.3 Findings

The findings of the study are presented according to the four specific objectives.

1. The study found that LaundryLogix needs structured information on users, transactions, machines, orders, and inventory to accurately model operations and support decisions.

2. The study found that applying the 4+1 Architectural View Model with UML diagrams produced a coherent cloud-based client-server architecture aligned with user needs, workflows, and deployment requirements.

3. The study found that the web-based POS and customer mobile app effectively transformed manual processes into integrated digital workflows that improved organization, transparency, and convenience.

4. The study found that LaundryLogix received positive USE ratings from customers, owner/staff, and laundry shop respondents, with most scores falling under "Agree," showing that the system is useful, easy to use, easy to learn, and satisfying.

7.4. Conclusions

Based on the findings, the following conclusions were drawn.

1. The researchers conclude that clearly defining information requirements for users, transactions, machines, orders, notifications, and inventory is essential for LaundryLogix to support efficient operations and accurately reflect Elyssa Laundry Center's workflow.
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2. The researchers conclude that a centralized, cloud-based architecture guided by the 4+1 Architectural View Model and UML diagrams provides a coherent structure aligned with the shop's functional, behavioral, development, and physical needs.

3. The researchers conclude that LaundryLogix features for POS machine management, stock monitoring, pricing, transaction history, analytics, pickup and delivery, reservations, digital payments, and the customer mobile app effectively convert manual processes into integrated digital workflows that improve transparency, organization, and convenience.

4. The study found that LaundryLogix received very favorable USE ratings from customers, owner/staff, and laundry shop respondents, with questions such as "It is easy to use," "It is user friendly," "Using it is effortless," and "I would recommend it to a friend" obtaining the highest mean scores in the scale, indicating that the system is useful, easy to use, easy to learn, and satisfying.

7.5 Recommendations

In light of these conclusions, the following recommendations are proposed.

1. It is recommended that future researchers and developers further analyze information requirements, especially user, transactional, and resource data, to refine LaundryLogix or similar systems.

2. It is recommended that future versions enhance the interface by organizing the mobile app into clearer tabs (Services and Prices, Machines, Self-Service vs. Full-Service), highlighting extras and reservation prices, and adding options like a second address field and automatic status labels for customer requests.

3. It is recommended that future iterations improve user experience by refining color coding, standardizing fonts and sizes, enlarging critical text where needed, considering a dark mode, and using less intrusive bottom notifications instead of center pop-ups.

4. It is recommended due to the responses of the lower-rated items in the USE Questionnaire, the researchers recommend improving the flexibility of the system and refining features that do not yet fully meet user needs, so that future versions of LaundryLogix can better support all customer and operational requirements.



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E.O.A
M.G.M.L
T.I.C.P



DEDICATION

This capstone project is dedicated to the researchers' beloved families, whose unwavering love, understanding, and support have been a constant source of strength and motivation throughout the completion of this study. Their encouragement during moments of difficulty and their belief in the researchers' abilities inspired them to persevere and accomplish this work.

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Above all, this work is dedicated to Almighty God, for the guidance, wisdom, and blessings that sustained the researchers in every stage of the project.

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Tristan

Gester



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Chapter 1

INTRODUCTION

Background of the Study

The laundry service industry plays an increasingly important role in modern society as lifestyles become busier and more urbanized. Daily routines are often dominated by activities such as school, office work, business operations, and commuting, leaving individuals and families with limited time for time-consuming household chores like washing, drying, and folding clothes. In this context, professional laundry services have emerged as a practical solution that allows customers to maintain cleanliness and hygiene while allocating their time to other priorities. For many students, employees, and small business owners, bringing clothes to a neighborhood laundry shop has become a regular part of weekly or even daily routines.

In the Philippine setting, the demand for convenient laundry services is particularly evident in urban and semi-urban communities where people often live in shared residences, boarding houses, or apartments with limited space for washing and drying clothes. In such environments, access to personal washing machines may be restricted, and



drying clothes outdoors can be difficult due to weather conditions or limited outdoor areas. Small and medium-sized laundry businesses, including neighborhood shops like Elyssa Laundry Center, respond to this need by providing accessible and reasonably priced services, helping residents maintain hygiene standards despite these constraints.

These local laundry shops do more than provide a service; they are part of the micro, small, and medium enterprise (MSME) sector that contributes to job creation and livelihoods in many communities. They typically employ attendants, cashiers, and sometimes delivery riders, and they often build long-term relationships with regular customers. Over time, the reliability and consistency of service become important factors that influence customer loyalty. When laundry operations are well-managed, customers tend to recommend the shop to friends and family, further strengthening its place in the local economy.

However, operating a laundry business involves more complexity than might be apparent at first glance. A single laundry shop may handle numerous orders daily, each with different customers, laundry weights, service preferences, deadlines, and special instructions. Staff must ensure that



each order is correctly tagged, processed, and returned to the right customer on time. They must also manage the allocation of washing machines and dryers, monitor inventory levels of consumable supplies, handle payments, and keep records for financial tracking. When these tasks are managed manually, the risk of errors and inefficiencies increases.

In many small laundry businesses, operations are still supported by traditional, paper-based methods. Customer information and service details are recorded in notebooks or on printed forms, receipts are handwritten, and service fees are calculated using simple calculators. Inventory is often checked by looking at containers and shelves to estimate whether detergent or fabric conditioner is running low. While such practices are familiar and require no specialized training or equipment, they are also prone to mistakes, inconsistencies, and information loss. Over time, these limitations can hinder the ability of the business to grow and respond to increasing customer demand.

The continued reliance on manual processes leads to recurring operational problems. Misplaced or damaged logbooks, receipts, and tags can make it difficult for staff to verify the details and status of a particular



order when customers return to claim their laundry. If two or more customers have similar names or drop-off schedules, confusion can arise, potentially leading to incorrect orders being released. Manual computations of service fees can result in undercharging or overcharging, especially when staff must consider various add-ons, surcharges, or promotions while attending to several customers at once. These issues create an environment where errors are more likely and where customer trust can easily be affected.

From a business perspective, inconsistent record-keeping and error-prone processes can also limit the ability of owners to understand the true performance of their shops. If transaction data are incomplete, inaccurate, or scattered across multiple notebooks, it becomes difficult to calculate actual daily or monthly sales, evaluate the profitability of certain service types, or determine whether price adjustments are needed. Similarly, without reliable records of inventory usage, owners may not know whether they are overstocking or understocking supplies, leading either to unnecessary expenses or stockouts that disrupt operations.

The literature on Point of Sale (POS) systems suggests that digital tools can help resolve many of these



challenges. Omega POS (2024) notes that modern POS systems are designed to help businesses process transactions more efficiently, maintain accurate inventory records, and improve customer satisfaction by reducing errors and delays. In service-oriented businesses such as laundries, POS systems can compute billing automatically based on standard rules, store customer information systematically, and generate digital receipts that are easier to track and retrieve than handwritten ones (Omega POS, 2024). By automating repetitive tasks, POS systems can also free staff to focus more on customer service.

Extenda Retail (2024) emphasizes that POS systems function not only as cash registers but as comprehensive platforms that organize information about products, services, and customers in a structured and accessible way. Their adaptability allows businesses to configure system modules according to their specific operational requirements (Extenda Retail, 2024). In a laundry context, this may include defining services such as wash only, wash and dry, wash-dry-fold, pressing, and express processing, as well as specifying pricing rules for each service. With this configuration, staff no longer need to memorize complex pricing schemes, and calculations become more



consistent and reliable.

Modern POS solutions often incorporate multiple modules that together support different aspects of business operations. Common modules include sales processing, inventory management, reporting, customer management, and sometimes employee management. In a laundry setting, a POS system can be extended with machine management features that track which washers and dryers are in use, and for which orders, at any given time. This integrated approach reduces the need for separate tools or informal tracking methods, such as sticky notes or verbal reminders, which can be easily overlooked or lost.

Beyond operational support, POS systems are increasingly recognized for their role in enabling data analytics. Krunchbox (2025) explains that POS data can be collected and analyzed to discover patterns in sales and customer behavior, such as which products or services are most popular, how purchasing changes across seasons, and which days or times are busiest. For a laundry shop, these insights might show that customers prefer to drop off laundry on certain days, that express services are in higher demand during certain weeks, or that particular types of detergent are frequently selected as add-ons



(Krunchbox, 2025). With this information, owners can plan staffing schedules, machine maintenance, and promotional strategies more effectively.

Resonai (2025) discusses the application of predictive analytics in retail environments, where historical data are used to anticipate future trends in customer behavior and inventory needs. In laundry operations, predictive analytics can support forecasting of how much detergent, fabric conditioner, and other consumables will be required over specific periods, based on past transaction patterns and seasonal fluctuations (Resonai, 2025). This predictive capability helps ensure that a laundry shop like Elyssa Laundry Center maintains adequate stock without over ordering, thereby optimizing cost management. Agatić, Tijan, Hess, and Jugović (2021) reinforce this view by showing, in the context of logistics, that advanced data analytics improves demand forecasting and resource planning. These findings, while not specific to laundries, highlight the potential benefits of applying analytics concepts in service businesses that must manage fluctuating demand and finite resources (Agatić et al., 2021).

Despite the documented advantages of POS systems and analytics, many small laundromats remain hesitant to adopt



them. Curbside Laundries (2024) notes that some laundromat owners perceive digital systems as too expensive, too complex, or both. Subscription fees, proprietary hardware, and specialized installation services can make commercial POS solutions appear out of reach for small operators who have limited budgets and minimal in-house technical expertise (Curbside Laundries, 2024). Owners may also worry that staff will struggle to learn new systems or that the transition might disrupt existing workflows. These concerns can delay or prevent digitalization even when manual processes are clearly inefficient.

Commercial POS products like Laundry Boss illustrate the advanced capabilities that larger laundromats can obtain when they invest in specialized systems. Laundry Boss (2025) describes features such as remote monitoring of machines, centralized control, detailed reporting, and integration with payment terminals and customer loyalty programs. While these features are powerful, they are typically geared toward laundromats that operate at a scale where such investments are justified by transaction volume and revenue (Laundry Boss, 2025). For small, independent shops, however, the costs and complexity of acquiring and maintaining such systems may outweigh perceived benefits,



leading them to continue relying on paper-based methods and simple calculators despite their limitations.

This situation illustrates a gap between high-end commercial solutions and the everyday realities of small laundry businesses, which require tools that are both affordable and aligned with their specific workflows. Research has therefore turned toward web-based management systems that can be run using standard, widely available hardware and accessed through web browsers. Nurhayani, Danov, Saputra, and Sari (2025) present a web-based laundry management system that automates administrative tasks, supports report generation, and manages customer data. Their implementation demonstrates that even smaller laundry businesses can benefit from digital systems that do not require costly proprietary hardware or complex installations (Nurhayani et al., 2025).

Arianto and Karyanto (2024) propose another web-based service management information system for a laundry business, using the Rapid Application Development method to quickly build and refine a solution tailored to actual operational needs. Their work shows that iterative development and close engagement with users can produce systems that match workflows more closely and are easier



for staff to adopt (Arianto & Karyanto, 2024). These studies suggest that web-based systems can provide a middle ground: more structured and powerful than paper-based methods, but less complex and expensive than some commercial POS packages.

In addition to the technical design of such systems, user experience and communication features are crucial. Kalo and Amron (2023) emphasize a user-centric approach to transforming laundry services, where system design is driven by the needs and behaviors of both staff and customers. They highlight that customers value transparency about service status, clear time estimates, and convenient access to information about their laundry (Kalo & Amron, 2023). This suggests that features such as order tracking, notifications, and easy-to-read dashboards are not optional extras but essential components of modern laundry service systems.

Zaad Platforms (2025) reinforce this perspective by discussing the key characteristics of effective POS systems for laundry operations. They identify real-time status visibility, intuitive interfaces, and comprehensive reporting as critical features that enable laundries to manage operations efficiently while keeping customers



informed (Zaad Platforms, 2025). When staff can see the status of every machine and order at a glance, they can make better decisions about which orders to prioritize, which machines to allocate, and how to respond to unexpected issues. When customers receive timely updates, they are less likely to experience confusion or frustration.

Visual analytics contributes to these capabilities by presenting complex data in forms that are easier to interpret than raw tables or logs. Lu, Yafeng, and colleagues (2021) explain that visual representations of data such as charts and graphs, support users in quickly understanding patterns and making decisions. In a laundry context, dashboards that show daily sales, machine utilization, and inventory trends can help owners identify anomalies, confirm assumptions, and monitor performance over time without needing advanced analytical training (Lu et al., 2021). When combined with interactive features, visual analytics can further support exploration and deeper insight into operational data.

Permana, Pramudito, and Sulaeman (2025) demonstrate how an intuitive web-based POS information system can significantly improve operational efficiency in a retail



context by standardizing processes and making information readily accessible to users. Although the study focuses on retail, the same principles apply to laundry services, which involve repeated transactions, inventory management, and the need for timely reporting. When processes are standardized through a system like a web-based POS, staff are less likely to deviate from preferred procedures, and management gains greater visibility and control over operations (Permana et al., 2025).

More broadly, Christian (2025) frames digital transformation in the laundry industry as a key strategy for achieving both efficiency and profitability. They argue that implementing digital solutions allows laundries to reduce operational errors, speed up service delivery, optimize resource use, and provide better customer experiences. Additionally, digital transformation opens opportunities for innovation, such as introducing new service models, marketing channels, or value-added features that would be difficult to manage using purely manual systems (Christian, 2025). For small enterprises, this transformation can be vital in keeping pace with evolving customer expectations and staying competitive in local markets.



Within this wider technological and industry context, Elyssa Laundry Center exemplifies the situation of many small laundry shops that continue to rely on manual, paper-based systems. Its operations involve recording customer details in notebooks, computing prices using calculators, tracking machine usage informally, and monitoring inventory through visual inspection. While these practices have allowed the business to operate, they also expose it to the same challenges described in the literature: risk of misplaced records, inconsistent pricing, limited visibility into operational performance, and difficulty providing timely and accurate information to customers.

For example, when multiple customers drop off laundry during peak hours, staff must juggle several tasks at once accepting clothes, weighing them, computing fees, issuing receipts, and assigning machines. Under such conditions, the likelihood of skipping a step or making a computation error increases. If a logbook is misplaced or a handwritten entry is unclear, staff may not immediately know which clothes belong to which customer, or whether a particular order has been fully paid. Over time, these small issues accumulate, consuming staff time and potentially leading to conflicts or dissatisfaction.



The absence of systematic data also limits the owner's ability to make strategic decisions. Without a digital record of transactions, it is difficult to determine precise daily or monthly sales, identify trends, or evaluate which services generate the most revenue. Without structured inventory data, it is hard to know exactly how quickly detergent, fabric conditioner, or packaging materials are being consumed or how frequently purchases are made. These gaps reduce the capacity of the owner to plan, budget, and optimize the use of resources.

Moreover, customer communication in such a setup is often limited to verbal promises and, at most, manual phone calls or text messages initiated by staff. Customers may be told to return at a certain time, but if they arrive earlier, they may find that their laundry is not yet ready. If they arrive later, their laundry may occupy limited storage space longer than necessary. In the absence of a systematic notification mechanism, customers may have to rely on guesswork or repeated inquiries, which can be inconvenient.

In response to these challenges and guided by the literature on POS systems, web-based management solutions, data analytics, visual analytics, and digital



transformation, the researchers proposed LaundryLogix, a digital Point of Sale (POS) system tailored specifically for Elyssa Laundry Center. LaundryLogix is conceptualized as a web-based platform that centralizes key operational functions, transaction processing, machine management, inventory monitoring, sales reporting, analytics dashboards, and customer notifications into a single integrated system.

The design of LaundryLogix draws on the advantages of POS technologies discussed by Omega POS (2024) and Extenda Retail (2024), applying them to the particular context of a small laundry business. It also reflects principles derived from research on web-based laundry management systems (Nurhayani et al., 2025; Arianto & Karyanto, 2024), user-centered service design (Kalo & Amron, 2023), effective POS features for laundries (Zaad Platforms, 2025), visual analytics (Lu et al., 2021), and intuitive web-based POS information systems (Permana et al., 2025). In essence, LaundryLogix aims to translate theoretical and practical insights from these studies into a concrete, context-appropriate solution for a single laundry shop.

In its initial conceptualization, LaundryLogix included SMS-based notifications, where the system would



send text messages to customers once their laundry was ready for pickup. As the project evolved, the researchers refined this design by replacing SMS notifications with in-app notifications delivered through a dedicated customer mobile application connected to the web-based POS. This shift aligns with the user-centric vision of Kalo and Amron (2023), who emphasize improved transparency and communication, and with Zaad Platforms' (2025) emphasis on real-time status visibility. It also reduces dependency on external SMS gateways and recurring messaging costs, which can be significant for small enterprises.

By integrating automation, analytics, visual dashboards, and in-app notifications, LaundryLogix aims to support Elyssa Laundry Center in transitioning from manual, paper-based processes to a more efficient and data-driven model of operation. The system is designed not only to minimize errors and speed up individual transactions but also to provide the owner with actionable information about the business as a whole. Through this transformation, Elyssa Laundry Center can align itself with the broader movement toward digitalization in the laundry industry, as discussed by Christian (2025), while still respecting its scale, resources, and local context.



In a wider perspective, the issues observed at Elyssa Laundry Center are not isolated but are representative of the broader challenges faced by many micro and small enterprises that provide services in traditional, face-to-face environments. Small service businesses often emerge from family initiatives or local entrepreneurial efforts and grow organically without the benefit of formal systems design or information management planning. As customer demand increases, existing manual processes are frequently stretched beyond their limits, leading to ad hoc solutions that may temporarily address problems but fail to provide a stable foundation for long-term growth.

The literature on POS systems and digital transformation demonstrates that structured, technology-supported processes can significantly improve reliability and scalability in such settings. Omega POS (2024) and Extenda Retail (2024) emphasize that the shift from manual to digital transaction handling is not merely a matter of replacing notebooks with computers; it involves rethinking how information flows through the business, from customer intake to payment to reporting. By formalizing this flow within a POS system, businesses reduce their dependence on individual memory and informal practices, making operations



less vulnerable to staff turnover, absences, or human error (Omega POS, 2024; Extenda Retail, 2024).

In addition, the integration of data analytics as discussed by Krunchbox (2025), Resonai (2025), and Agatić et al. (2021) invites small enterprises to adopt a more evidence-based approach to decision-making. Instead of relying on intuitive judgments or rough estimates, owners can use historical data to understand patterns in customer behavior, resource consumption, and financial performance. For example, a laundry owner who has access to clear visualizations of weekly and monthly sales can identify which periods require more staff on duty, or whether specific promotional campaigns correlate with increased demand for certain services (Krunchbox, 2025; Lu et al., 2021). This level of insight is difficult to achieve using manual records scattered across multiple notebooks.

Studies of web-based laundry systems by Nurhayani et al. (2025) and Arianto and Karyanto (2024) show that even in specialized service environments, it is possible to design systems that align closely with actual workflows. Their findings suggest that when systems are developed with direct input from end users and refined through iterative feedback, adoption barriers can be significantly reduced.



Staff members are more likely to embrace a system whose screens and processes reflect the steps they already follow, but in a clearer and more structured form (Nurhayani et al., 2025; Arianto & Karyanto, 2024).

The user-centric emphasis of Kalo and Amron (2023) and the focus on relevant features by Zaad Platforms (2025) further support the notion that system success depends not only on technical correctness but also on how well the system supports both staff and customers in their daily interactions. A laundry system that provides real-time information about order status, clear pricing, and transparent timelines is better positioned to meet customer expectations. At the same time, staff benefit from interfaces that minimize cognitive load, reduce the number of steps required to complete tasks, and help them avoid mistakes (Kalo & Amron, 2023; Zaad Platforms, 2025).

Visual analytics, as discussed by Lu et al. (2021), contributes an additional dimension to this user-centric vision by making complex operational patterns easier to understand. For small laundry owners who may not have formal training in statistics or data science, visual dashboards can act as an accessible entry point into data-driven management. When sales, machine usage, and inventory



data are displayed in simple charts and graphs, owners can more readily spot anomalies, opportunities, and long-term trends that might otherwise remain hidden in raw data tables (Lu et al., 2021).

The application of intuitive web-based POS designs in other sectors, such as the retail case examined by Permana et al. (2025), offers a template for how similar principles can be brought into laundry operations. Their study shows that when systems are built with simplicity and clarity in mind, they can facilitate smoother day-to-day operations, better information control, and more reliable reporting (Permana et al., 2025). Translating these lessons to the context of Elyssa Laundry Center implies that a carefully designed web-based POS can help the shop move beyond basic logbooks and calculators toward a more professional and robust management approach.

Christian (2025) frames such changes in terms of digital transformation, emphasizing that the adoption of digital tools is not just a technical upgrade but a strategic shift in how laundry businesses operate and compete. For small enterprises, digital transformation can open doors to new service models, such as integrating online booking, loyalty features, or more personalized



communication. Even when the initial implementation focuses only on core operational functions such as transaction handling, machine management, and basic analytics, it lays the groundwork for future enhancements and innovation (Christian, 2025).

Despite the availability of these concepts and tools, there remains a gap in how they are applied in practice within small, single-branch laundry shops in local communities. Much of the commercial POS landscape, including products like Laundry Boss, targets larger laundromats that have the resources to invest in advanced hardware, extensive training, and ongoing subscriptions (Laundry Boss, 2025). On the other hand, many academic prototypes focus on general models or specific case studies that may not fully reflect the constraints of micro-enterprises. This creates a space for research projects like this capstone, which aim to design and evaluate systems tailored specifically to a single small laundry shop, using technologies and design principles that are realistic for that context (Nurhayani et al., 2025; Arianto & Karyanto, 2024; Curbside Laundries, 2024).

Elyssa Laundry Center provides an appropriate setting for this investigation because it exhibits many of the



characteristics discussed in the literature: reliance on manual recording, vulnerability to human error, limited formal data analysis, and a growing need for more efficient and transparent operations. By focusing on a concrete, real-world case, this study can explore not only the technical feasibility of implementing a web-based POS system with analytics and in-app notifications, but also the practical challenges of adoption, usability, and alignment with existing workflows.

The LaundryLogix system is designed to directly address these identified needs and gaps. Its core modules covering transaction processing, machine management, inventory monitoring, reporting, analytics dashboards, and customer notifications are grounded in established concepts from the POS and information systems literature. At the same time, its specific configuration and user interface are intended to reflect the day-to-day realities of Elyssa Laundry Center rather than the needs of a generic or large-scale laundromat. In this way, LaundryLogix represents a localized application of broader digital transformation ideas discussed by Christian (2025), Omega POS (2024), Extenda Retail (2024), and others.

In addition, the refinement from SMS-based



notifications to in-app notifications through a dedicated customer mobile application illustrates how the project adapts to both literature and context. User-centric studies emphasize the importance of timely, clear, and cost-effective communication with customers (Kalo & Amron, 2023; Zaad Platforms, 2025). By choosing in-app notifications, the system can provide richer, more flexible updates without incurring per-message costs typical of SMS gateways, an important consideration for small enterprises managing tight budgets. This design decision aligns with the goal of enabling sustainable, long-term use of the system.

Overall, the extended discussion of literature and context leads to a clear articulation of the research gap that this study seeks to fill. While prior works have shown the potential of POS systems, data analytics, web-based laundry management, user-centric design, and digital transformation to improve service operations, there is a lack of implemented and evaluated systems specifically crafted for small, single-branch laundry shops like Elyssa Laundry Center that integrate these elements in a comprehensive yet practical way. By focusing on the design, development, and evaluation of LaundryLogix, this study



intends to respond to that gap and to provide a model that can inform similar initiatives in other small laundry enterprises.

In response to these challenges, the researchers proposed LaundryLogix, a digital Point of Sale (POS) system tailored to Elyssa Laundry Center. Conceived as a web-based platform, it centralizes core functions such as transaction processing, machine management, inventory monitoring, and sales reporting. By digitalizing the operations of a small laundry enterprise through affordable, context-appropriate tools, the project supports Sustainable Development Goal 9 (Industry, Innovation, and Infrastructure) by fostering innovation and improving access of small businesses to modern technologies.

Company Profile

Elyssa Laundry Center is a small, independently owned laundry service business located in Loakan Road, Baguio City, Philippines. It primarily serves residents, students, and workers living and working in nearby communities, many of whom have limited space, time, or equipment for doing laundry at home. The shop offers basic wash, dry, and fold services, as well as options for special handling based on fabric type and customer preference. Its operations are



largely walk-in based, with customers dropping off and picking up their laundry at the physical shop.

In response to these challenges and guided by the literature on POS systems, web-based management solutions, data analytics, and digital transformation, the researchers proposed LaundryLogix, a digital Point of Sale (POS) system tailored specifically for Elyssa Laundry Center.

LaundryLogix is conceptualized as a web-based platform that centralizes core functions such as transaction processing, machine management, inventory monitoring, and sales reporting into a single integrated system. By modernizing the operations of a micro-scale laundry enterprise through affordable, context-appropriate digital tools, the project supports Sustainable Development Goal 9 (Industry, Innovation, and Infrastructure), which promotes resilient infrastructure, innovation, and improved access of small enterprises to modern technologies..

Importance of the Study

The purpose of this study is to design, develop, and evaluate LaundryLogix, a web-based Digital Point of Sale (POS) system integrated with a customer mobile application, that will address the operational challenges of Elyssa Laundry Center by automating transactions, improving



monitoring of machines and inventory, and enhancing communication with customers through in-app notifications. The study seeks to provide a practical, user-centered solution that reflects the advantages of POS and analytics technologies discussed in prior research while remaining appropriate for the scale and context of a small laundry business (Omega POS, 2024; Extenda Retail, 2024; Krunchbox, 2025; Resonai, 2025; Christian, 2025).

The significance of the study can be viewed in terms of who will benefit from the developed system.

Owner of Elyssa Laundry Center. The owner of Elyssa Laundry Center shall benefit from LaundryLogix through reduced manual errors, standardized pricing, optimized machine usage, and more systematic inventory control, which together can support more efficient operations and better-informed business decisions.

Staff of Elyssa Laundry Center. The staff of Elyssa Laundry Center shall benefit because the system consolidates transaction encoding, machine assignment, and inventory tracking into a single interface, simplifying daily workflows, minimizing repetitive computations, and making records easier to manage.

Customers of Elyssa Laundry Center. The customers of



Elyssa Laundry Center shall benefit by receiving clearer and timelier information about their laundry status, more organized transaction handling, and fewer errors, resulting in a smoother and more transparent service experience.

Researchers. The researchers shall benefit by gaining hands-on experience in applying agile development, user-centered design, architectural modeling, and usability evaluation in a real-world small-business context, thereby strengthening their technical and research competencies.

Future Researchers. Future researchers shall benefit by using this project as a reference for enhancing similar systems, extending features such as advanced analytics or IoT-based machine monitoring, or adapting the approach to other service-oriented small and medium enterprises.

Objectives of the Study

The main objective of this study is to design and develop LaundryLogix, a Digital Point of Sale System that streamlines service transactions for Elyssa Laundry Center. Specifically, it will aim to answer the following:

1. to identify the information requirements of the system;
 2. to determine to architectural framework of the system;
-



3. to identify the features of the system; and
4. to measure the extent of usability of the system.

Definition of Terms

This section clarifies how key technical and operational terms are used in the context of the LaundryLogix project so that readers can interpret the succeeding chapters consistently and accurately.

Analytics Dashboard. The Analytics Dashboard is the visual interface that presents key indicators such as total sales, machine utilization, and inventory status through charts and graphical summaries to support data-driven decisions.

Architectural Framework. The Architectural Framework is the structured way of describing a software system using multiple complementary views, such as logical, process, development, and physical perspectives, to show how components interact and how the system is organized from design to deployment.

Cloud-Based Architecture. Cloud-Based Architecture is the deployment setup where the LaundryLogix system is hosted on online servers and accessed through a web browser, reducing the need for on-site hardware and supporting access from different devices within the shop.



Customer Mobile Application. The Customer Mobile Application is the mobile client that connects to LaundryLogix and allows customers to view their order status and receive in-app notifications, replacing traditional SMS-based updates.

Digital Point of Sale (POS) System. The Digital Point of Sale System is the computerized system used by Elyssa Laundry Center to record transactions, manage machine assignments, track inventory, generate reports, and support decision-making, replacing manual, paper-based transaction handling.

Digitalization. Digitalization is the shift from manual, paper-based processes—such as handwritten logs and calculator-based billing—to computerized workflows using LaundryLogix, with the goal of reducing errors, improving record-keeping, and enhancing service transparency.

In-App Notifications. In-App Notifications are messages sent from LaundryLogix to the customer mobile application to inform customers about important events such as order acceptance, washing completion, and readiness for pickup.

Information Requirements. Information Requirements are the specific data elements and records that a system must



capture, store, and process for example, users, transactions, machines, orders, notifications, and inventory so that it can accurately represent operations and support decision-making.

LaundryLogix. LaundryLogix is the web-based POS system, specifically designed for Elyssa Laundry Center, that integrates transaction processing, machine management, inventory control, reporting, analytics, and customer notifications.

Machine Management Module. The Machine Management Module is the component of LaundryLogix that allows staff to assign washers and dryers to customer orders, monitor their status, and manage service parameters such as cycle type and duration.

Sales Reporting. Sales Reporting is the functionality that generates summaries of transactions over defined periods (for example, daily, weekly, monthly), which the owner can use to analyze business performance.

Usability. Usability is the extent to which LaundryLogix can be used by Elyssa Laundry Center staff and customers to achieve their goals effectively, efficiently, and with satisfaction in the intended context of use.

Web-Based System. Web-Based System refers to software,



like LaundryLogix, that runs through a web browser over the internet rather than being installed as a standalone desktop application, enabling easier deployment, updates, and access for the laundry shop.





Chapter 2

METHODOLOGY

This chapter presents the procedures on how this study obtained the data from various sources.

Software Development Methodology

The researchers reviewed both traditional and modern software development methodologies before selecting an approach.

The Waterfall model proposes a linear sequence of phases, requirements analysis, system design, implementation, testing, deployment, and maintenance where each phase is completed before the next begins, making it appropriate when requirements are stable and well understood; however, it becomes less suitable when requirements are expected to change or when continuous user feedback is necessary, as is the case for LaundryLogix, which was developed in close coordination with a single laundry shop whose needs and ideas evolved during the project, particularly in relation to the customer interface and notification mechanisms, making a strictly linear approach too rigid. In response, the researchers turned to Agile methodologies, which emphasize iterative development, frequent feedback, and continuous improvement,



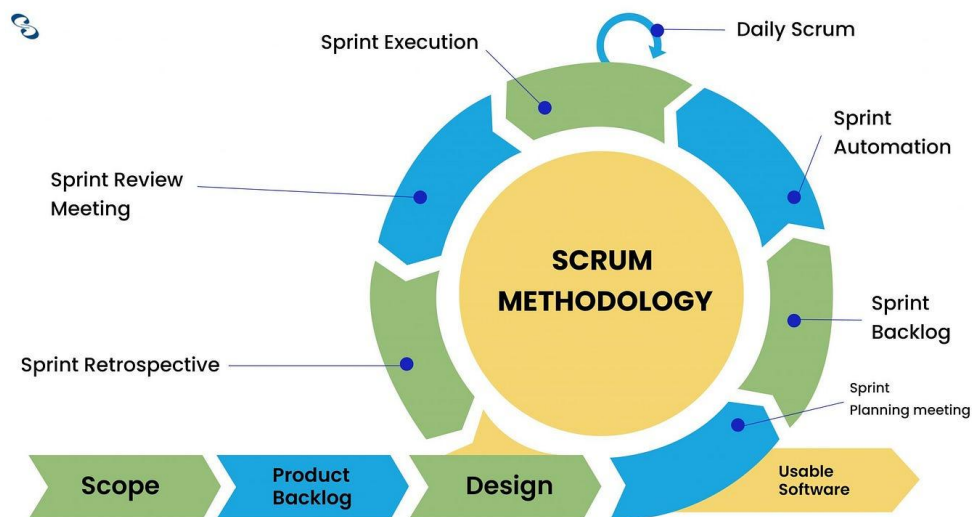
and reviewed several approaches: Feature-Driven Development (FDD), which focuses on designing and building by feature and is effective in large, complex projects where features can be clearly identified and scheduled; Scrum, which organizes work into time-boxed sprints with defined roles (Product Owner, Scrum Master, Development Team) and ceremonies (planning, daily stand-ups, reviews, retrospectives) and is well suited for projects that need iterative delivery and regular stakeholder feedback; Extreme Programming (XP), which highlights technical practices such as test-driven development, pair programming, and continuous integration to improve code quality; Lean, which concentrates on eliminating waste, optimizing flow, and delivering maximum value with minimal resources; Kanban, which uses a visual board to manage work in progress and emphasizes continuous flow rather than fixed-length sprints; Rational Unified Process (RUP), a structured, iterative methodology often applied in larger software projects with clearly defined roles and phases; and ASAP Methodology, commonly used in enterprise environments (such as SAP implementations) to accelerate and structure large-scale deployments, with Scrum ultimately serving as the primary reference for organizing

LaundryLogix development because of its balance between structure and flexibility.

After comparing these methodologies, Scrum was selected as the primary framework because it balances structure and flexibility, supports short iteration cycles, and encourages continuous feedback from stakeholders essential conditions for a capstone project with an actively involved client.

Figure 1

Scrum Development Methodology



The application of Scrum in the LaundryLogix development involved the following elements:

Product Backlog Creation. The team created a product backlog consisting of user stories derived from interviews and observations at Elyssa Laundry Center. These user



stories captured what different users needed to accomplish with the system, such as "As a staff, I want to record a new laundry transaction so that I can correctly compute the total payment" and "As a customer, I want to see my laundry's status on my phone so that I know when it is ready for pickup." This backlog served as the central list of requirements to be refined and implemented throughout the development process.

Sprint Planning. During sprint planning, the product backlog was prioritized based on importance, dependencies, and feasibility within the project timeline. Features were then grouped into sprints, with Sprint 1 focusing on the core POS transaction flow and basic database structure, Sprint 2 on machine assignment, inventory monitoring, and simple reporting, Sprint 3 on the customer mobile application user interface and in-app notifications, and Sprint 4 on refinements, usability improvements, and preparation for evaluation. This structure ensured that each sprint delivered a meaningful, testable increment of the system.

Daily (or Regular) Stand-ups. Short coordination meetings, conducted in person or online, were held regularly to monitor progress. In these stand-ups, each



team member briefly reported what they had completed, what they planned to do next, and any obstacles they were facing. This practice helped the team quickly identify bottlenecks, redistribute tasks when necessary, and maintain alignment on sprint goals.

Sprint Review. At the end of each sprint, a sprint review was conducted in which the team demonstrated completed features to the adviser and the Elyssa Laundry Center owner. Stakeholders provided feedback on the implemented functionalities, including suggestions such as adjusting labels, rearranging screen elements, or modifying notification timing. These comments were then translated into new or updated user stories in the product backlog and scheduled for future sprints.

Sprint Retrospective. After each sprint review, the team held a sprint retrospective to reflect on the development process itself. They discussed what went well, what did not go as planned, and what could be improved in terms of communication, task breakdown, and defining acceptance criteria. Lessons learned from each retrospective were applied to subsequent sprints to continuously improve the team's workflow and productivity.

By following Scrum, the team was able to iteratively



refine LaundryLogix and ensure that both the POS and mobile interfaces remained aligned with actual user needs.

Design Thinking

Design Thinking is a problem-solving approach that focuses on understanding user needs, defining key problems, generating creative ideas, and building and testing prototypes. In this study, it guided the development of LaundryLogix through the Empathize, Define, Ideate, Prototype, and Test phases, ensuring that both the POS system and the customer mobile application were grounded in the real context of Elyssa Laundry Center.

In the Empathize phase, interviews, observations, empathy maps, and a 5 Why Analysis were used to capture what the owner and staff say, think, do, and feel about daily operations. These activities revealed heavy reliance on manual records, difficulty tracking income, frequent risk of misplaced logs, time-consuming encoding, and stress during peak hours due to paper-based tracking of receipts, orders, and machine availability. The Empathy Maps and 5 Why Analysis figures originally illustrated these pain points and traced them to root causes such as low prioritization of technology, misconceptions about cost and complexity of digital tools, and lack of workflow redesign



and training.

During the Define phase, the insights from Empathize were synthesized into a clear problem statement focused on inefficient manual transaction handling and the lack of real-time monitoring in laundry operations. Owner and staff personas were created to summarize goals, frustrations, and needs, such as accurate income tracking, organized records, better visibility of business performance, faster customer service, and simpler machine monitoring. The persona figures previously visualized these characteristics and helped ensure that functional requirements stayed aligned with both management and frontline priorities.

In the Ideate phase, the researchers explored multiple solution concepts and evaluated them against feasibility, accessibility, and alignment with existing workflows. Using tools such as an Impact-Effort Matrix and a Business Model Canvas, the team prioritized high-impact features like a customer database, order tracking dashboard, inventory monitoring, and machine reservation over more complex, lower-priority enhancements. The Impact-Effort Matrix figure showed how features were grouped into quick wins and major projects, while the Business Model Canvas summarized key partners, activities, resources, value propositions,



customer segments, channels, relationships, costs, and potential revenue logic for LaundryLogix.

In the Prototype phase, high-fidelity mockups were produced for both the web-based POS and the customer mobile application, translating the defined requirements into concrete interface designs. The POS prototype consolidated customer details, service selection, machine assignment, inventory tracking, and payment summary into a single workflow, while the mobile prototype emphasized machine availability, reservations, and order status viewing with clear labels and status indicators. Previously, the prototype figures visually demonstrated how this interface elements replaced scattered manual records with organized digital flows for both staff and customers.

Finally, in the Test phase, the team conducted simulated transactions and user feedback sessions to evaluate the prototypes. Testing focused on validating automated computations, inventory alerts, reservation handling, and the synchronization of status updates and in-app notifications between the POS and the mobile app. Feedback from the owner, staff, and customers confirmed that the interfaces were intuitive, that machine assignment and reservation logic worked as intended, and that in-app



notifications were sufficient to keep customers informed about order and reservation status.

Scope and Delimitation

This section describes what is covered by the study and what is deliberately excluded.

In this phase of the research, the formal usability evaluation is concentrated on customers using the mobile application, while usability testing of the staff-facing POS interface is acknowledged as important but reserved for future work.

The system uses a cloud-based client-server architecture. The client side includes a Next.js and Tailwind CSS web POS and owner dashboard plus a customer mobile interface, while the backend uses Next.js API routes, Firebase Cloud Firestore, Firebase Authentication, and Genkit to centralize data and synchronize updates in real time. Integration with external accounting systems, third-party payment gateways, and IoT machines is excluded from the current scope.

The system covers staff-facing POS functions and customer-facing mobile features. The web POS handles customer registration, transaction encoding with automatic fee computation, machine assignment and status, inventory



tracking with alerts, basic sales reports, simple analytics, and role-based user management. The mobile app lets customers securely access their account, view active and past transactions with status, and receive in-app notifications such as when orders are ready for pickup.

Delimitations. The system is specifically designed and configured for a single laundry shop, Elyssa Laundry Center, and does not support multi-branch management. More advanced financial modules, such as comprehensive profit and loss statements or detailed accounting workflows, are beyond the scope of this version. The evaluation conducted in this study focuses on subjective usability, using the USE Questionnaire to capture customer perceptions, and does not measure objective performance indicators like exact task completion times or error rates. Consequently, the findings and conclusions are limited to the context of Elyssa Laundry Center and its customers, and any generalization to other businesses or settings should be approached with caution.

Data Gathering Techniques

Several data gathering techniques were employed to support the different stages of the study:

Interview. Semi-structured interviews were conducted



with the owner and selected staff. These interviews explored current operational practices, pain points in manual processes, expectations for a digital system, and constraints such as staff familiarity with technology and available devices.

Survey. To measure usability of the LaundryLogix mobile application, the researchers administered the USE Questionnaire by Lund, which evaluates four aspects: usefulness, ease of use, ease of learning, and satisfaction. The instrument was implemented via Google Forms, and responses were collected from 100 customers after they had used the mobile app.

Document Analysis. Existing shop documents such as manual logbooks, handwritten receipts, and simple inventory notes were examined. This analysis provided insights into the structure of data currently being recorded, the kinds of information often missing or unclear, and patterns in transaction records.

Sources of Data

The study drew on both primary and secondary data sources. The primary sources consisted of the owner and staff of Elyssa Laundry Center, who provided first-hand information on existing operational processes, challenges,



and system requirements through interviews and observations, the 20 laundry shops, as well as the 100 customers who used the LaundryLogix mobile application and completed the USE Questionnaire, thereby supplying direct usability feedback based on their experience with the system.

The secondary sources included existing operational documents from Elyssa Laundry Center, such as logbooks, receipts, and simple inventory notes, which helped the researchers understand current record-keeping practices and data structures, and published studies and online resources related to POS systems, Agile and Scrum methodologies, the USE Questionnaire, and data privacy particularly the Philippine Data Privacy Act of 2012 (RA 10173) which guided the design of both the system and the research procedures.

Software Development Tools

Development of LaundryLogix utilized a set of tools chosen for their suitability for web and mobile interfaces and their integration capabilities:

Next.js. Next.js was used to implement the web POS interface and backend API endpoints. Its support for server-side rendering and API routes facilitated efficient page rendering and cleaner separation between frontend and



backend logic.

Tailwind CSS. Tailwind CSS provided utility classes for rapid UI development, allowing the team to create responsive, consistent layouts without writing extensive custom styles. This was particularly useful for ensuring that the POS interface remained clear and usable in different screen sizes.

Genkit. Genkit was used to handle event-driven logic related to notifications. Changes in Firestore, such as updates to order status, triggered notification workflows that sent relevant in-app messages to customers.

Cloud Firestore. Stored structured data for customers, transactions, machine states, inventory items, and notification logs.

Firebase Authentication. Managed user identities and access control, distinguishing between staff and customer accounts.

Visual Studio Code. Served as the main integrated development environment (IDE) for writing and organizing project code.

Firebase Console. It was used to configure Firestore collections, security rules, and authentication settings.

PayMongo Payments API. LaundryLogix uses the PayMongo



Payments API to handle secure digital payments (e.g., cards and e-wallets) and to track payment status in real time within the POS and mobile application.

GitHub. GitHub provided version control, enabling the team to track changes, manage branches, and collaborate efficiently. It also served as a backup and coordination platform, ensuring that code history, issues, and feature updates were documented and synchronized across developers.



Chapter 3

FINDINGS OF THE STUDY

This chapter presents the findings of the study based on the data gathered, analyzed, and interpreted by the researchers in line with the specific objectives. It discusses the information requirements of LaundryLogix system, the architectural framework that supports these requirements, the system's implemented features on both the POS and mobile sides, and the results of the usability evaluation conducted among the customers of Elyssa Laundry Center.

Information Requirements of the System

To address the operational inefficiencies caused by manual, paper-based transaction handling at Elyssa Laundry Center, the researchers first identified the core information requirements needed to digitize the workflow. Through empathy mapping and problem definition sessions with the owner and staff, the team clarified which pieces of information must be captured, stored, and processed in order to support day-to-day operations as well as monitoring and decision-making.

User and Authentication Requirements. to ensure secure access and personalized service for both customers and



administrators. The system must maintain basic profile data such as full name, contact number, and email address, while also managing encrypted login credentials to protect customer histories and enforce role-based access levels that distinguish between the shop owner and laundry staff. By explicitly modeling these data elements, the system prevents unauthorized access and allows actions and transactions to be tied to specific accounts, which is vital for accountability and traceability in a service environment.

Service and Transactional Data. The backbone of the digital transition from physical logbooks to a structured database. For each service event, LaundryLogix needs to capture reservation timestamps, service types (such as wash only, dry only, wash-dry-fold, or express), weight measurements, and any special instructions provided by the customer. In addition, to support modern business operations, the system must process and record payment data, including transaction IDs, payment status (pending or paid), and the method of settlement, whether in cash or through an integrated digital payment gateway. This level of detail ensures that each transaction is documented consistently, enabling accurate billing, reporting, and



review for both day-to-day and long-term analysis across periods and services.

Asset and Machine Monitoring. To eliminate “first-come, first-served” confusion and unmanaged queues on the shop floor. Each washer and dryer requires an identity within the system, including a machine ID, unit type or load capacity, and real-time operational status (available, in-use, or under maintenance). By continuously tracking these attributes, the system can display updated machine availability to staff and customers, thereby reducing the likelihood of double booking, idle capacity, or bottlenecks during busy periods.

Order Tracking and Notification Logic. A critical information requirements identified by the researchers. Each order must have a documented status history that reflects its progress from “Received” to “Processing” and eventually to “Ready for Pickup,” with intermediate stages as needed. Associated with this status trail are notification triggers, which act as data points that signal the automated dispatch of alerts to the customer’s device when key changes occur. This combination of status history and event triggers addresses previous communication gaps by ensuring that both staff and customers have a shared, up-



to-date understanding of where a particular order is in the process.

Inventory and Resource Management. For the administrative oversight of consumable supplies. The system maintains records of stock levels for essential items such as detergents, fabric softeners, and disinfectants, along with minimum stock thresholds that define when an item is considered low. When stock falls below these thresholds, the system can generate alerts for the admin, preventing unexpected supply shortages that might otherwise halt operations or reduce service quality. Taken together, these information requirements provide a structured foundation that allows LaundryLogix to replace fragmented manual records with an integrated, digital representation of Elyssa Laundry Center's operations.

Architectural Framework of the System

Based on the identified information requirements and the technical constraints of a small-to-medium enterprise, the researchers determined that a centralized, cloud-based hybrid architecture would be the most effective framework for LaundryLogix.

To describe this architecture in a systematic and stakeholder-oriented manner, the study adopted the 4+1

Architectural View Model. This model organizes the system into four primary views the logical view for functional requirements, the process view for dynamic behavior, the development view for code and component organization, and the physical view for deployment to hardware, unified by a fifth view centered on representative use cases.

Figure 2

4+1 Architectural Framework

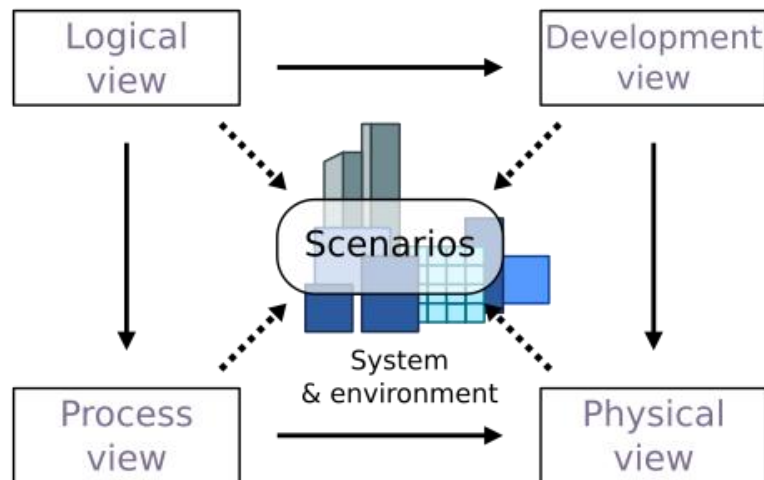


Figure 2 presents the 4+1 Architectural Framework applied to LaundryLogix, showing how the logical, process, development, and physical views are all anchored on key use cases to form a coherent system description. By using this framework, the researchers aligned the software architecture with the operational needs of Elyssa Laundry Center, from core transaction flows to hardware deployment.

Use Case Diagram. A diagram that is a behavioral Unified Modeling Language (UML) diagram that illustrates the high-level functional requirements of a system. It provides a visual representation of the interactions between external entities, known as "actors," and the various processes or "use case" operating within the system boundary. The primary purpose of this diagram is to establish the system's scope and demonstrate what the system does from an end-user perspective, without detailing the internal programming logic.

Figure 3

LaundryLogix Use Case Diagram

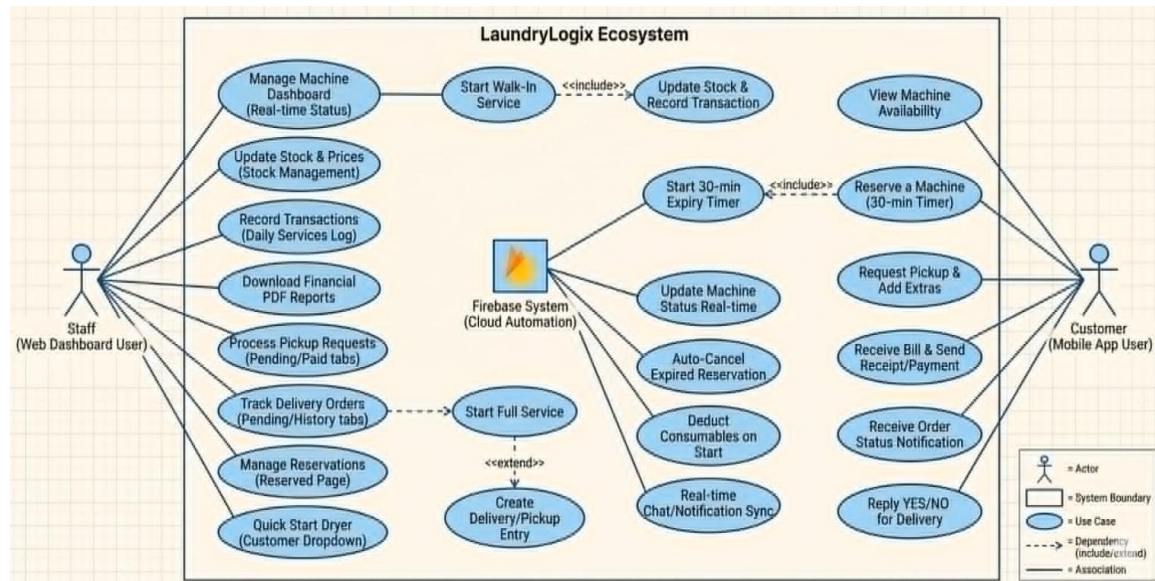


Figure 3 shows the LaundryLogix Use Case Diagram, which clearly separates the functionalities available to the two main actors: the Customer and the Staff/Owner. The



figure indicates that the Customer, primarily using the mobile application, can sign up or log in, update their profile, check machine status, reserve machines, select payment methods, complete payments, track laundry status, and receive notifications, thereby covering the entire customer journey from initial account creation to order completion. The figure also illustrates that the Staff/Owner, representing the laundry shop personnel using the web-based dashboard, is responsible for logging in, managing user accounts, managing inventory items, assigning machines to reservations, updating laundry status, and generating sales reports, with explicit include and extend relationships specifying how shared subprocesses (such as updating machine status whenever a machine is assigned) are reused across multiple use cases.

Logical View. The logical view describes the system's functional structure from the perspective of end-users and analysts. For LaundryLogix, it models core entities such as Customer, StaffOwner, Machine Reservation, Payment, LaundryStatus, and Inventory, and specifies how they relate to support tasks like managing accounts, recording reservations, tracking machine states, updating order status, and monitoring consumable stocks. This view ensures



that the system structure faithfully reflects the information requirements and workflows of Elyssa Laundry Center without yet addressing timing or hardware deployment details.

Class Diagram. A structural diagram that serves as the blueprint for the system's object-oriented design. It shows the static structure by identifying core classes, their attributes and operations, and the relationships between them, and is essential for translating business rules into a logical database schema.

Figure 4

LaundryLogix Class Diagram

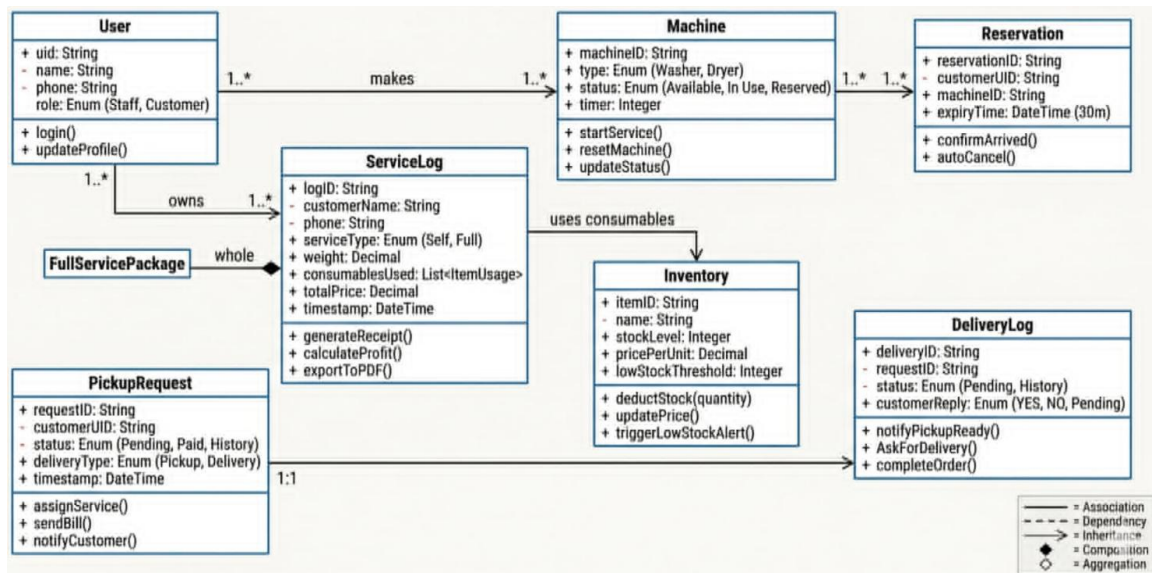


Figure 4 presents the LaundryLogix Class Diagram, which organizes the system into seven primary classes: Customer, StaffOwner, Machine, Reservation, Payment, LaundryStatus,



and Inventory. Customer and StaffOwner share common attributes (name, phone, email, password) but implement different methods: Customer includes operations like `reserveMachine()`, `makePayment()`, and `trackStatus()`, while StaffOwner supports `manageUsers()`, `manageInventory()`, and `generateReport()`, reflecting their distinct roles. Machine defines properties such as `machineID`, `type`, `capacity`, and `status`; Reservation links Customer and Machine with date, time, and reservation state; LaundryStatus records the current and historical states of orders; Payment holds transaction details; and Inventory monitors consumable items and low-stock conditions. Together, these attributes, methods, and relationships demonstrate how the logical structure of the system supports consistent data handling across both the POS and mobile application.

Process View. The process view describes the dynamic behavior of LaundryLogix and how operations unfold over time. It models scenarios such as customers logging in, checking machine availability, making reservations, paying, and monitoring order status, as well as staff or owner actions that update laundry status and confirm reservations. By detailing the ordering of operations, interactions, and concurrency between components, this view helps ensure the



system can handle multiple simultaneous users and transactions without losing consistency or responsiveness.

Sequence Diagram. A sequence diagram is an interaction UML diagram that details how operations are carried out over time in a system. It captures the dynamic behavior of a specific scenario by showing the chronological sequence of messages exchanged between actors and system objects along vertical lifelines, with time progressing from top to bottom. In this project, the sequence diagram is utilized to understand and document the flow of control necessary to complete a single business process, such as creating a transaction, assigning a machine, or processing a pickup request in LaundryLogix.

Figure 5

LaundryLogix Sequence Diagram

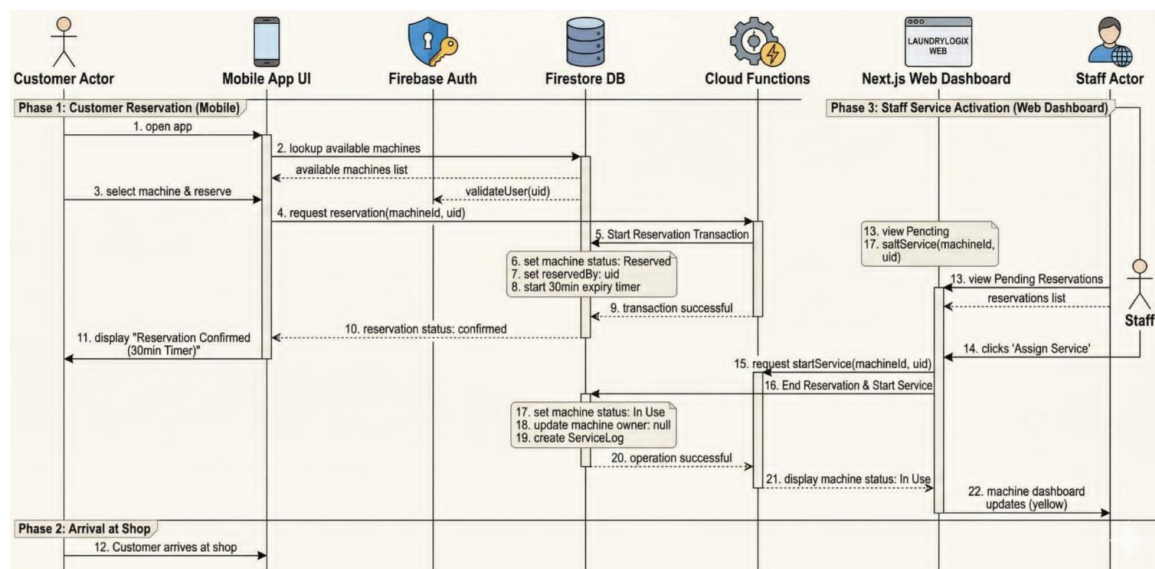




Figure 5 presents the LaundryLogix Sequence Diagram for the machine reservation and status tracking workflow, tracing the interaction from the moment a customer logs into the system up to the receipt of status notifications. The figure shows that the process begins when the Customer sends login credentials to the System, which then communicates with the Database to verify the information before granting access. The figure further illustrates how, after logging in, the Customer requests a list of available machines, prompting the System to query the Database for current machine states and return the results. Once the Customer selects a machine and issues a `reserveMachine()` call, the System requests payment details, processes the payment, and records the Reservation and Payment information back into the Database. The figure also demonstrates how the System then notifies the Staff/Owner of the new booking, and how subsequent updates made by the Staff/Owner to the `LaundryStatus` are saved to the Database and propagated to the Customer through notifications. By visualizing this end-to-end interaction, the figure confirms that the process view supports a closed communication loop and maintains transparency across all parties.



Development View. The development view, sometimes called the implementation view, organizes LaundryLogix into software modules and packages that support effective coding, testing, and maintenance. For this system, the view identifies packages such as Customer UI, Staff/Owner UI, User Management, Transaction Processing, Inventory Management, Database Access, and Payment Integration. It clarifies which classes and components belong to each package, and how these packages depend on one another. This view is particularly important for the development team because it guides source code organization, promotes modularity, and reduces coupling between unrelated concerns.

Package Diagram. A package diagram is a structural UML diagram used to organize the various elements of a complex system into higher-level groupings, known as packages. It visualizes the dependencies between these different packages, helping developers understand the macroscopic architecture of the software and how major subsystems rely on one another. In practice, a package diagram is highly effective for managing system complexity by separating distinct functional areas into modular layers or subsystems (for example, presentation, business logic, and data

access), which promotes modularity, encapsulation, and clearer impact analysis when changes are made.

Figure 6

LaundryLogix Package Diagram

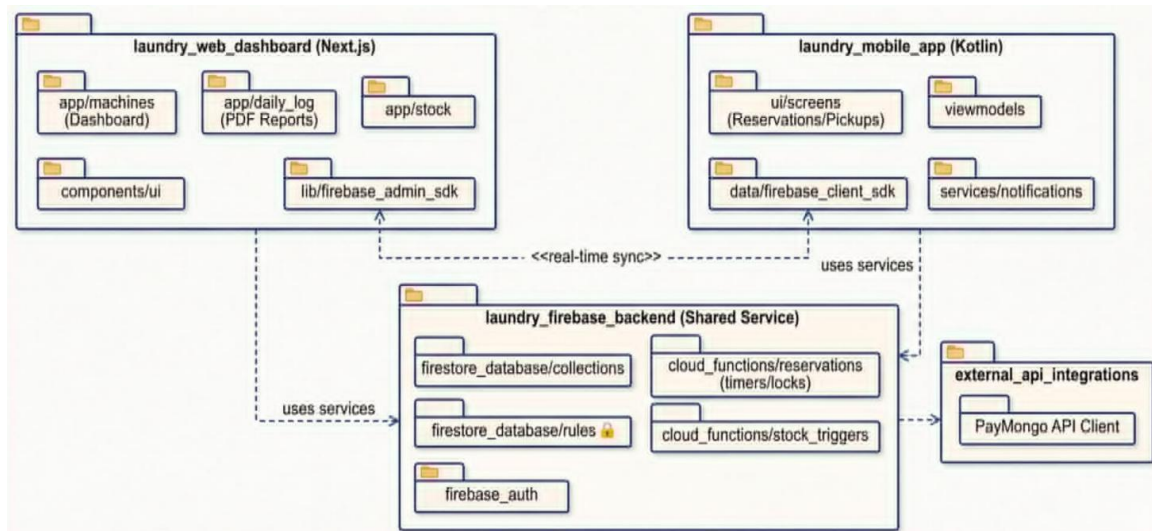


Figure 6 displays the LaundryLogix Package Diagram, which divides the application into the Customer UI and Staff/Owner UI at the presentation layer and business logic packages such as User Management, Transaction Processing, and Inventory Management at the core layer, all converging on a centralized Database package. The figure shows how both UI packages depend on User Management to handle authentication and profile operations, while Transaction Processing encapsulates logic for reservations, machine allocation, and payment calculations, and Inventory Management focuses on tracking and updating consumable stock. The figure also indicates a specialized dependency



from Transaction Processing to an external Payment Gateway package, which handles secure communication with digital payment services. By organizing components into these packages and clearly showing their dependencies, the figure demonstrates how the development view promotes modularity, maintainability, and clear separation of concerns within the LaundryLogix codebase.

Process View. The process view addresses the dynamic behavior of LaundryLogix and illustrates how operations unfold over time. In this view, scenarios such as a customer logging in, checking machine availability, making a reservation, paying, and then monitoring order status are expressed in terms of message flows and interactions among system components. It also depicts how staff/owner actions such as updating laundry status or confirming reservations trigger changes that propagate back to customer interfaces. By detailing concurrency, ordering of operations, and communication between components, the process view makes sure that the system can handle multiple simultaneous users and transactions without sacrificing consistency or responsiveness.

Deployment Diagram. A deployment diagram is a structural UML diagram that visualizes the physical

execution environment of the system and how software is actually deployed. It maps software artifacts such as executables, web services, configuration files, and databases onto hardware nodes like web servers, application servers, smartphones, and office computers where they will reside and run at runtime. This diagram is crucial for understanding network topology, hardware requirements, and communication paths between nodes, helping designers reason about performance, reliability, and security in distributed setups such as a cloud-hosted POS backend connected to staff terminals and customer mobile devices.

Figure 7

LaundryLogix Deployment Diagram

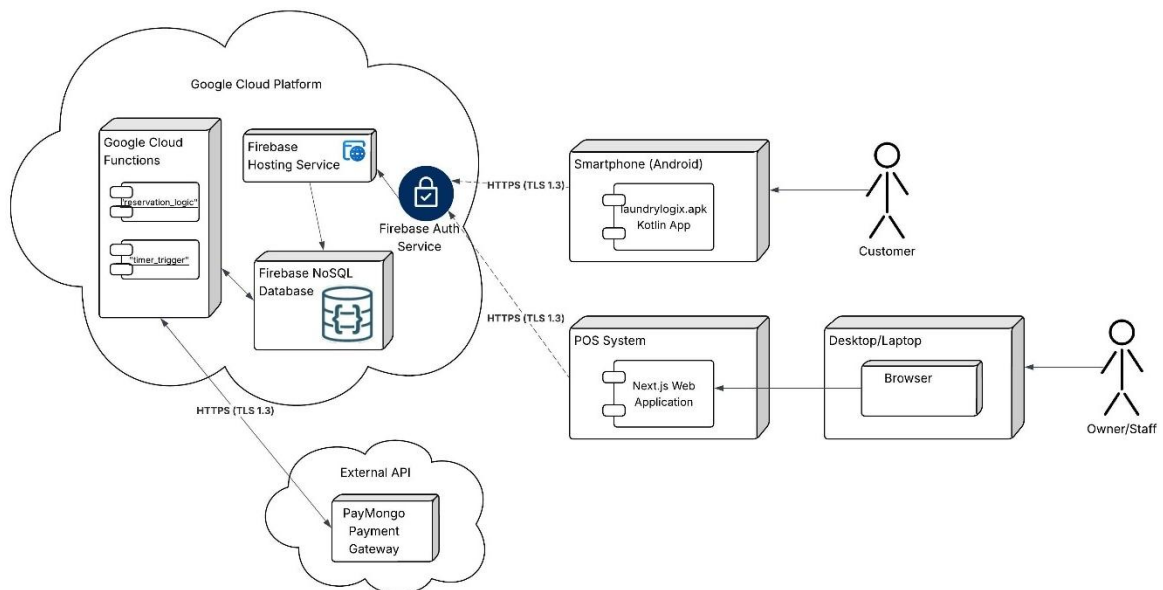


Figure 7 illustrates the LaundryLogix Deployment Diagram, mapping the customer mobile application to a User



Device (mobile phone) that hosts the compiled app, and mapping the staff/owner POS and dashboard functionalities to an Admin/Staff Device (PC) accessed through a standard web browser. The figure shows both client nodes communicating over the internet via HTTP/HTTPS with a centralized Cloud Server node. Within this cloud environment, the figure indicates a Web Server (implemented using web technologies) responsible for servicing requests and executing business logic, and a Database Server (using a document-oriented or relational database) responsible for storing and retrieving persistent data. The figure further highlights a connection from the Web Server to an External Payment Gateway node, which processes digital payments securely. By laying out these nodes and communication links, the figure demonstrates that the physical view of LaundryLogix is designed to be scalable, secure, and aligned with the actual operating conditions and constraints of Elyssa Laundry Center.

Features of the System

The researchers successfully developed LaundryLogix to serve as an integrated digital ecosystem, transforming traditional laundry operations into a modernized process. The system encompasses the following core features:



Figure 8

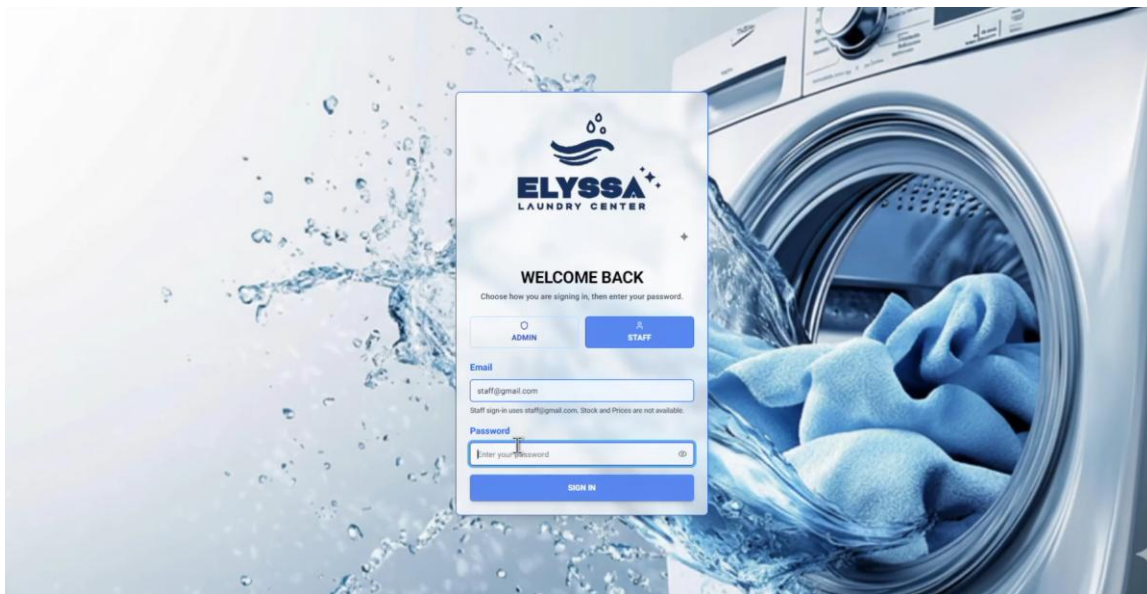
Web-based POS Login Page

Figure 8 demonstrates the precise manner in which the LaundryLogix platform deploys a highly secure, centralized authentication gateway as its primary line of defense, effectively mandating the verification of valid user credentials prior to granting permission to any operational Point of Sale (POS) functions. This strict barrier prevents unauthorized access and ensures that sensitive day-to-day operational data, financial ledger records, and private customer information remain accessible exclusively to verified individuals. Furthermore, the system incorporates Role-Based Access Control (RBAC) to establish a clear operational distinction between administrative owners and standard staff members, automatically restricting floor



staff to essential transactional utilities while seamlessly unlocking specialized data analytics reports and system configuration modules for owners. Consequently, this architectural design successfully enforces secure, role-specific access permissions, resulting in a much more structured, transparent, and organized daily workflow for the business.

Figure 9

Web-based POS Machine Management



Figure 9 illustrates the Machine Management dashboard of the LaundryLogix web-based POS, where washers and dryers are organized into clearly labeled categories and represented as individual cards with real-time status indicators. The figure shows that each machine card displays the machine identifier and a color-coded status



label, with green indicating "Available" in the example view, allowing staff and the owner to quickly assess overall capacity at a glance. The figure also highlights the use of a dark-themed interface to improve visual contrast and user focus, while a side navigation panel provides direct access to other operational modules such as stock monitoring, daily service logs, pickup requests, and reservations. By presenting all machines in one consolidated view and eliminating the need for physical inspection or manual lists, the figure demonstrates how this feature digitizes and streamlines the core monitoring and allocation workflow for daily laundry operations.

Figure 10

Web-based POS Stock Management

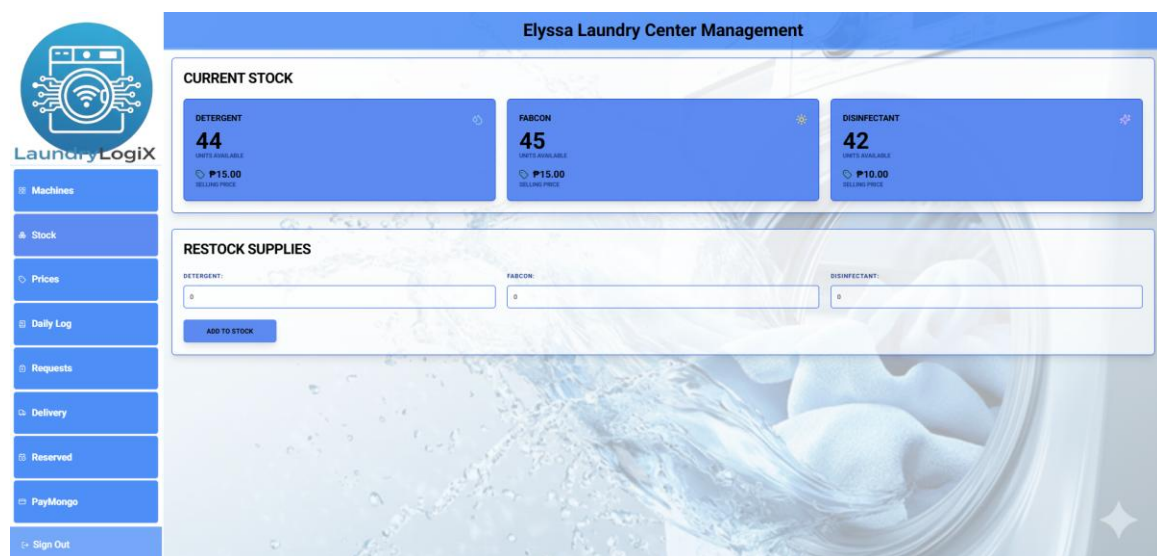


Figure 10 presents the comprehensive Stock Management



module, which features a highly detailed, visually organized inventory table populated with essential consumable items such as various detergents, fabric conditioners, and bleaching agents, each accompanied by its corresponding product image, exact name, and real-time remaining quantities to ensure rapid identification and reduce human error. As clearly illustrated in the figure, each row within the inventory matrix is equipped with an interactive "Added Stock" input field paired with a dedicated "Update" button, allowing authorized personnel whether operational staff or the business owner to immediately record and log newly delivered supplies directly into the system. Furthermore, the figure makes it apparent that once an update is submitted, the backend architecture automatically recalculates and stores the newly adjusted totals, seamlessly providing an instantaneous and up-to-date view of the entire inventory ecosystem without the need for error-prone manual ledger entries or time-consuming physical counts of containers. By completely replacing traditional, ad hoc tracking methods with a highly structured digital record, the figure effectively demonstrates how this real-time feature minimizes the risk of costly stockouts, delivers actionable



data to support wiser purchasing decisions, and ultimately contributes to much smoother, uninterrupted daily business operations.

Figure 11

Web-based POS Prices and Information Management

MANAGE PRICES & INFO

SERVICES

SELF SERVICE WASH 80	FULL SERVICE WASH 220	RESERVATION FEE 30
DRYER (1 HOUR) 90	DRYER (EXTRA 30 MINS) 50	FOLDING SERVICE 15
DELIVERY SERVICE 50	SELF PICK-UP 0	LAUNDRY PICKUP 50
EXCESS WEIGHT (PER KG) 30		

CONSUMABLES

DETERGENT 15	FABCON 15	DISINFECTANT 10
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SHOP CONTACT DETAILS

PRIMARY CONTACT 09557255873	SECONDARY CONTACT 09054340569
---------------------------------------	---

* THESE NUMBERS WILL BE DISPLAYED ON THE MOBILE APP'S SERVICES AND INFO SCREEN.

SAVE ALL CHANGES

Figure 11 shows the Prices and Information Management interface, where the shop owner configures standard service rates and maintains key business details such as the official contact number. Labeled input fields allow the owner to define or update prices for each service type so that changes are automatically reflected in POS computations and in the prices shown to customers, removing the need for staff to memorize complex pricing rules or rely on outdated printed lists. By centralizing price and



information management in a single, secure screen, this feature supports consistency, accuracy, and quick adaptation to changing costs or business policies.

Figure 12

Web-based POS Transaction History

TRANSACTION HISTORY

Saturday, April 11, 2026 at 01:17 AM

Filter by customer...

CUSTOMER	MACHINE	TIME	TYPE	SERVICE	STOCKS USED & DETAILS	REVENUE	PROFIT	PAYMENT
mark go	Washer #2	4/9/2026, 10:13 AM	MOBILE PICKUP	Full Service - Mobile Pickup (7kg, 45min)	Service Charge	P220.00		
					Detergent (x1)	INCLUDED		
					Fabcon (x1)	INCLUDED		
					Disinfectant (x1)	INCLUDED		
					Dryer Service (1 Hour)	INCLUDED		
					Fold	INCLUDED		
					Extra Detergent (x1)	P15.00		
					Extra Fabcon (x1)	P15.00		
Enrico abenes	Washer #1	4/9/2026, 10:04 AM	WALK-IN	Self Service Wash (7kg, 45min)	Extra Disinfectant (x1)	P10.00		
					Delivery Service	P50.00		
					P310.00	P256.00	DIGITAL	
Enrico abenes	Washer #1	4/9/2026, 10:02 AM	WALK-IN	Self Service Wash (7kg, 45min)	Service Charge	P80.00		
					Detergent (x2)			
					P80.00	P80.00	CASH	
					P80.00	P80.00	CASH	

Figure 12 presents the Transaction History module, which functions as a digital ledger of completed laundry services for Elyssa Laundry Center. The figure shows a tabular layout capturing key details for each transaction, including customer name, machine used, transaction type (walk-in or reservation), total amount paid, and date of service, with a search bar, date filters, and a “View Details” button to quickly locate and inspect specific records. By replacing handwritten logbooks with this structured, filterable ledger, the feature improves record-keeping, supports audits and reporting, and enhances



transparency in the shop's financial and operational history.

Figure 13

Web-based POS Performances and Forecasting Analytics



Figure 13 shows the Performance and Forecasting Analytics dashboard, where historical and projected business metrics are visualized using combined bar and line charts to give the owner a clearer view of both past performance and expected future trends. Key indicators such as Total Sales, Net Sales, and Average Transactions per Day are displayed at the top, while filters for daily, monthly, and yearly views allow the owner to explore trends, identify peak periods, and observe seasonal patterns over time. Forecasting curves are overlaid on the charts to



indicate likely future performance based on past data, helping the owner plan staffing, manage inventory budgets, and evaluate business growth using evidence-based insights.

Figure 14

Web-based POS Request Pickups

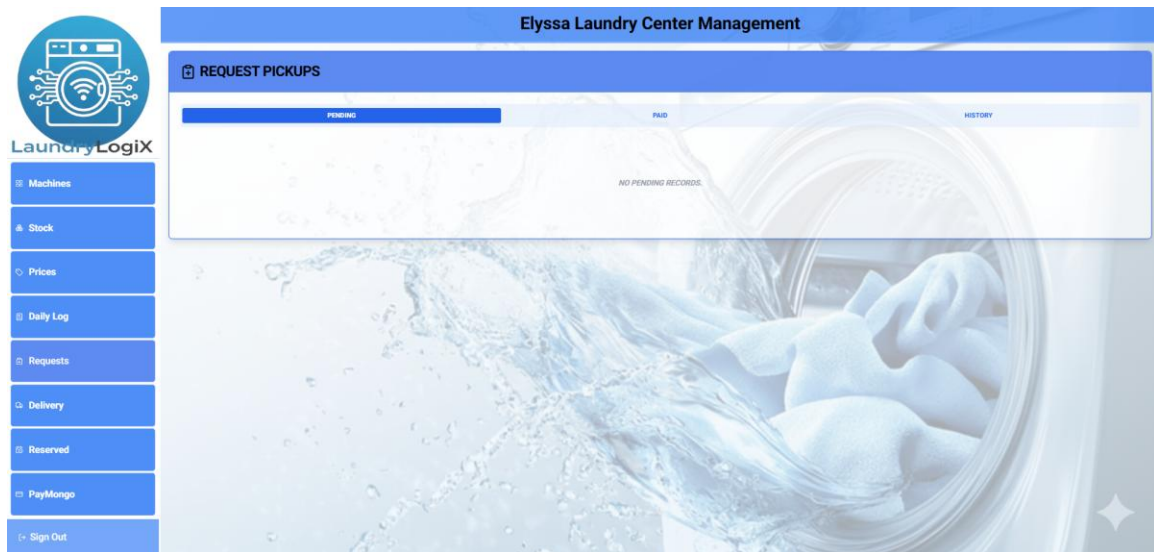


Figure 14 displays the LaundryLogix Pickup Request interface within the customer mobile application, designed to support customers who prefer off-site collection of their laundry. The figure shows that the screen includes a date picker for choosing a pickup date and a location input field linked to a map, enabling customers to specify their exact pickup location with precision. The figure also shows a "Submit Request" button that, when tapped, transmits the request data to the POS system. By capturing pickup details in this structured digital format and synchronizing them



with the Staff/Owner dashboard's pickup module, the figure demonstrates how the feature eliminates the need for informal coordination through calls or messages, simplifies scheduling for the shop, and provides customers with a more convenient, modern way to request services.

Figure 15

Web-based POS Delivery/Pickup Management

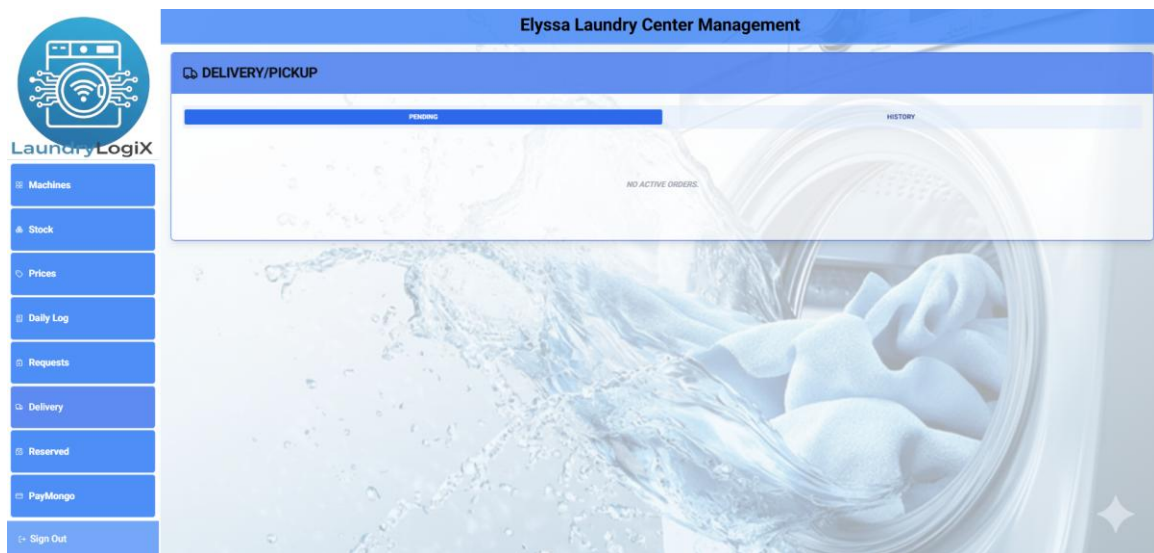


Figure 15 shows the Delivery/Pickup Management module of the LaundryLogix web-based POS, giving staff and the owner a centralized view of all active orders that require either delivery or pickup completion. Each entry displays key details such as customer information and service type, along with a "Done" button that staff can click once a delivery or pickup has been fulfilled, keeping the list of pending tasks up to date and reducing the chance of



overlooked orders. By consolidating logistical information into a single, easy-to-monitor interface, this feature improves coordination, minimizes missed or delayed handovers, and enhances the efficiency of end-to-end service fulfillment.

Figure 16

Web-based POS Machine Reservation Management

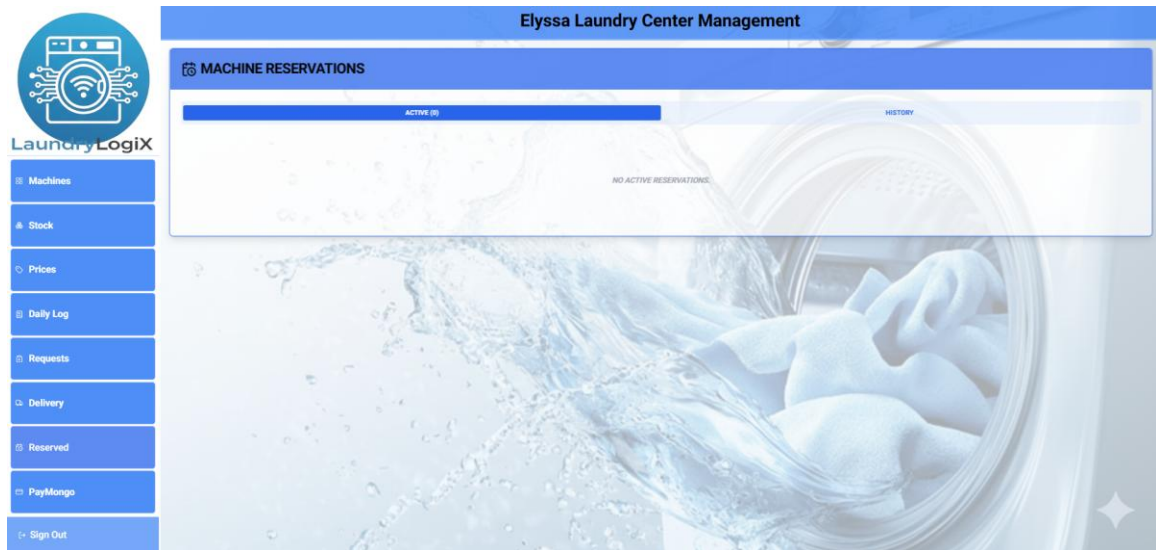


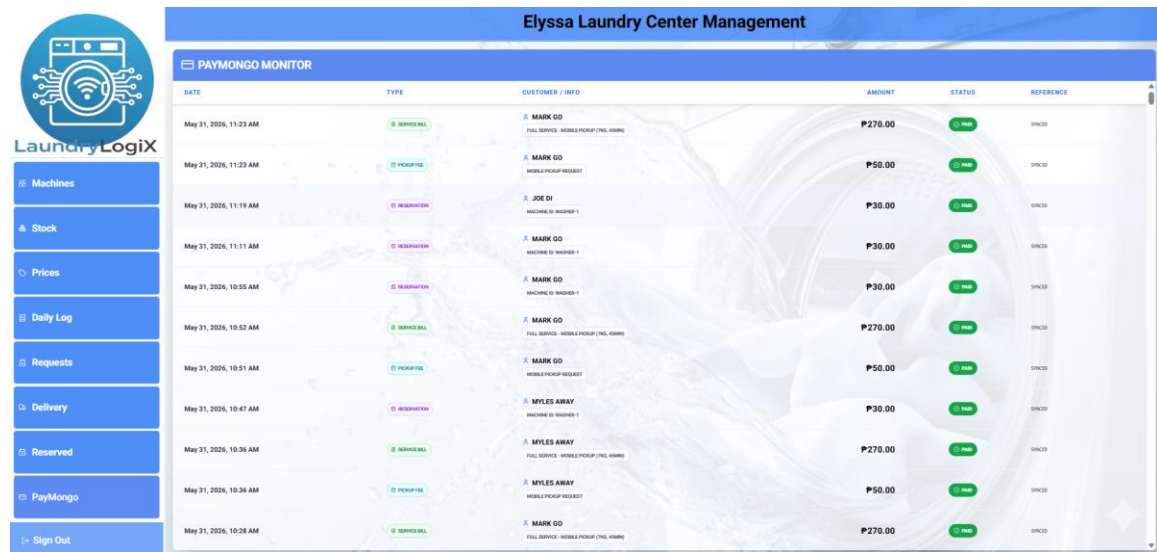
Figure 16 shows the Machine Reservation Management interface of the LaundryLogix POS, which consolidates all machine reservations submitted through the customer mobile application into a single table. Each row lists the reservation date, customer name, and assigned washer or dryer, along with “Accept” and “Cancel” buttons that let staff confirm or reject requests based on current shop capacity. By structuring reservations this way and tying



them to actual machine availability, the feature helps prevent double-booking, improves machine utilization, and balances walk-in and reserved usage.

Figure 17

Web-based POS PayMongo Monitoring Page



DATE	TYPE	CUSTOMER / INFO	AMOUNT	STATUS	REFERENCE
May 21, 2026, 11:23 AM	SERVICEBILL	MARK GO FULL SERVICE - MOBILE PICKUP (PHIL. KASINO)	P270.00	PAID	SPK020
May 21, 2026, 11:23 AM	PICKUPFEE	MARK GO MOBILE PICKUP REQUEST	P50.00	PAID	SPK020
May 21, 2026, 11:19 AM	RESERVATION	JOE DI MACHINE ID: M000001-1	P30.00	PAID	SPK020
May 21, 2026, 11:11 AM	RESERVATION	MARK GO MACHINE ID: M000001-1	P30.00	PAID	SPK020
May 21, 2026, 10:55 AM	RESERVATION	MARK GO MACHINE ID: M000001-1	P30.00	PAID	SPK020
May 21, 2026, 10:52 AM	SERVICEBILL	MARK GO FULL SERVICE - MOBILE PICKUP (PHIL. KASINO)	P270.00	PAID	SPK020
May 21, 2026, 10:51 AM	PICKUPFEE	MARK GO MOBILE PICKUP REQUEST	P50.00	PAID	SPK020
May 21, 2026, 10:47 AM	RESERVATION	MYLES AWAY MACHINE ID: M000001-1	P30.00	PAID	SPK020
May 21, 2026, 10:36 AM	SERVICEBILL	MYLES AWAY FULL SERVICE - MOBILE PICKUP (PHIL. KASINO)	P270.00	PAID	SPK020
May 21, 2026, 10:36 AM	PICKUPFEE	MYLES AWAY MOBILE PICKUP REQUEST	P50.00	PAID	SPK020
May 21, 2026, 10:28 AM	SERVICEBILL	MARK GO FULL SERVICE - MOBILE PICKUP (PHIL. KASINO)	P270.00	PAID	SPK020

Figure 17 illustrates the PayMongo Monitoring Page integrated into the LaundryLogix POS, which serves as the main dashboard for tracking digital payment transactions. The interface presents a table listing each payment's reference ID, amount, date, and status (for example, "Paid" or "Pending") and is updated in near real time, giving staff and the owner immediate visibility into whether a customer's digital payment has been processed successfully. By making it easy to cross-check app-based payments against corresponding laundry service records, this feature



strengthens financial control, reduces discrepancies, and supports accurate revenue reconciliation.

Figure 18

Web-based POS Staff Interface



Figure 18 shows the Staff Interface of the LaundryLogix web-based POS, which presents a streamlined and simplified version of the main owner dashboard that is specifically tailored to optimize frontline store operations. As illustrated in the figure, this interface prioritizes the essential functions required for daily service execution, including real-time machine management, transaction encoding, and the tracking of delivery or pickup confirmations. The visual layout clearly demonstrates that administrative modules such as inventory stock management and service price configurations are



completely omitted and inaccessible from this view. This design directly reflects the system's role-based access control architecture, which restricts high-level backend adjustments exclusively to the owner, thereby allowing frontline staff to concentrate entirely on executing day-to-day customer tasks accurately, efficiently, and without the risk of accidental systemic errors.

Figure 19

Mobile Application Dashboard

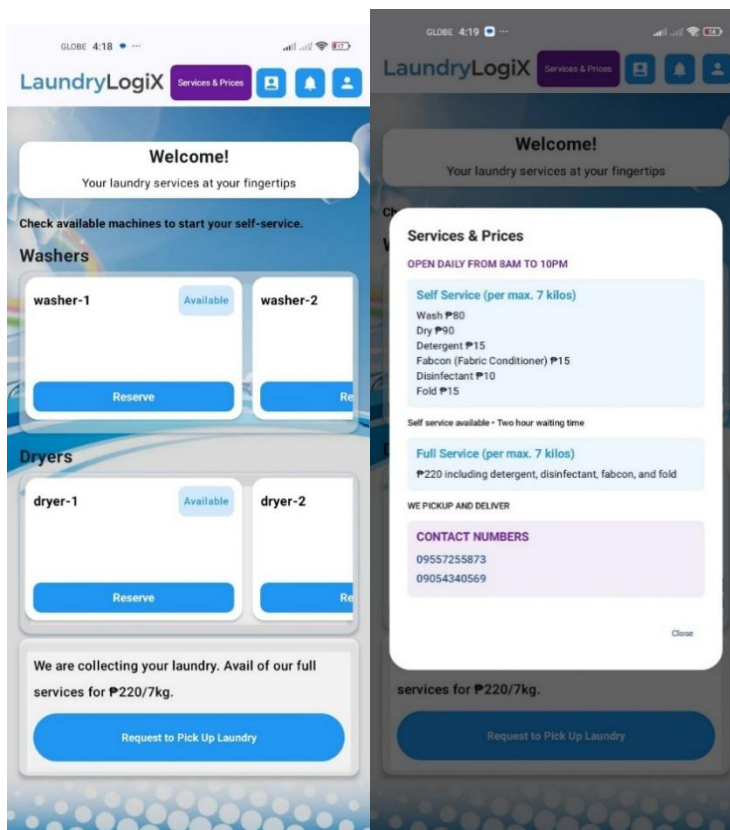


Figure 19 shows the primary dashboard of the LaundryLogix mobile application, which serves as the main interface for customers of Elyssa Laundry Center. The



screen features a user-centric layout with a promotional carousel for announcements or offers, a grid of key services (such as self-service, drying, and pickup or delivery), and a live Machine Availability indicator that displays the number of available machines before customers decide to visit or reserve. By integrating these elements into a single, intuitive screen, the dashboard helps customers quickly understand their options, choose the most convenient time and service, and reduce unnecessary waiting or overcrowding at the shop.

Figure 20

Mobile Application Machine Reservation

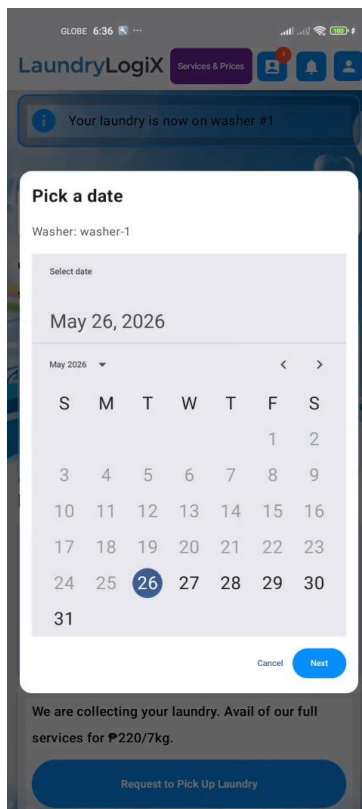




Figure 20 shows the Machine Reservation interface within the LaundryLogix mobile application, which provides customers with a highly intuitive, self-service digital gateway to reserve a specific washing machine or dryer for their preferred date and time slot. As illustrated in the figure, the screen guides users through a highly structured, step-by-step workflow where they can seamlessly input their desired schedule, select from a dynamically updated list of available appliances based on real-time shop capacity, and transparently view the corresponding reservation fee before tapping a prominent "Proceed to Payment" button to finalize the booking. Furthermore, the interface operates on a rigorous validation framework that instantly cross-references requests against the live database to eliminate overlapping bookings, double-reservations, or incomplete entries. By empowering clients to secure their machinery well in advance, this digital feature helps customers plan their laundry visits with pinpoint precision, drastically reduces in-shop congestion and physical queuing times, and provides management with a far clearer, highly predictable schedule of machine utilization throughout the day. Consequently, this seamless integration of mobile scheduling bridges the gap between



customer convenience and operational efficiency, transforming a traditionally unpredictable chore into a highly optimized, stress-free retail experience for both the business and its patrons.

Figure 21

Mobile Application Request Pickup

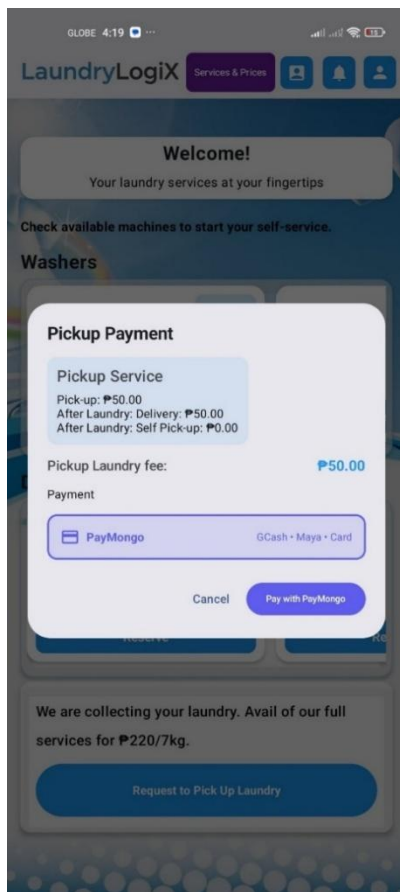


Figure 21 shows the Request Pickup interface of the LaundryLogix mobile application, designed for customers who want the shop to collect their laundry from home or another location. The screen provides form fields for selecting a pickup date and specifying an address, along with a "Submit



Request” button that sends the details to the POS system, with validation helping ensure that required information is complete before submission. By digitizing pickup requests, this feature streamlines communication, reduces manual coordination, and creates a clearer, more reliable record of pickup commitments for both customers and the laundry center.

Figure 22

Mobile Application Receipts

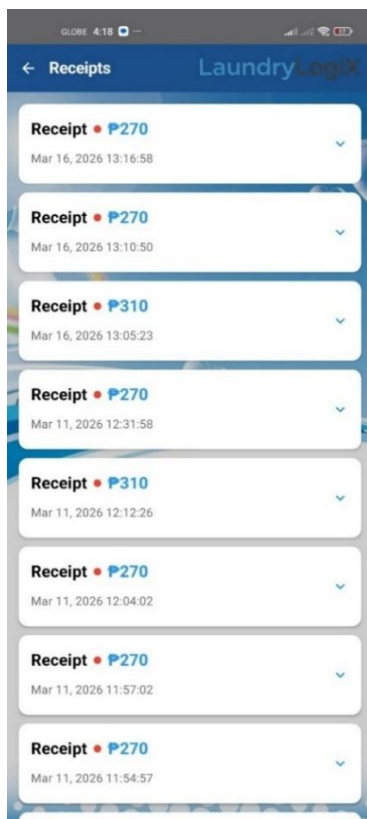


Figure 22 shows the digital Receipts section of the LaundryLogix mobile application, which gives customers an organized record of their past transactions in place of



traditional paper receipts. The interface lists receipts with details such as reference ID, total amount, and transaction status, and tapping an entry opens a more detailed view of the services rendered and payment method used. By centralizing receipts in one accessible location, this feature helps customers conveniently track expenses, provides a reliable record for questions or disputes, and enhances financial transparency.

Usability Testing of the System

To determine how effectively LaundryLogix supports its intended users in real operating conditions, the researchers conducted a structured usability evaluation with actual customers and with the owner and staff of Elyssa Laundry Center. This evaluation focused on key usability dimensions usefulness, ease of use, ease of learning, and satisfaction to capture not only whether the system works, but also whether users perceive it as practical, intuitive, and pleasant to use. By gathering quantitative ratings through a standardized questionnaire, the study aimed to provide evidence-based support for claims that LaundryLogix is acceptable and ready for deployment in a small laundry business setting.

The target population for the usability test included



two main groups: (1) customers who transact with laundry shops and use LaundryLogix in the context of their laundry transactions, and (2) the shop owner and staff who use the web-based POS to encode transactions, assign machines, and monitor inventory. In principle, every eligible customer and staff member could have been invited to participate; however, time constraints, availability of respondents, and the need for manageable data collection made it necessary to draw a smaller but representative sample. To address this, the researchers relied on Slovin's formula to estimate an appropriate sample size for customers while including all available owner/staff as part of a complete enumeration.

Slovin's formula is commonly used in survey research to determine the minimum sample size required when the total population size is known and the researcher wants to limit the margin of error to a specific level. This approach is suitable for exploratory usability studies like this one because it provides a simple yet systematic way to balance precision with the practical realities of data collection.

Where: n = sample size

N = population size

e = margin error



$$n = \frac{N}{1 + Ne^2} \quad n = \frac{13,837}{1 + 13,837(0.10)^2} = 99.28$$

n = 100 Customer Respondents

For the customer group, the population consisted of individuals who transact with laundry shops and are potential users of LaundryLogix for managing their laundry orders. Because this population is larger and more diverse than the internal staff, the researchers used a more relaxed 10% margin of error to balance accuracy with practical constraints such as time and respondent availability. Using this margin of error in Slovin's formula produced the required customer sample size, which then served as the basis for the number of questionnaires distributed, ensuring that the results would reasonably reflect customer perceptions while keeping data collection manageable.

To measure usability in a structured and comparable way, the researchers used the USE (Usefulness, Satisfaction, and Ease of use) Questionnaire as the primary instrument. The questionnaire was organized into four sections corresponding to the dimensions of Usefulness, Ease of Use, Ease of Learning, and Satisfaction, with several statements under each dimension that respondents rated using a seven-point Likert scale from 1 (Strongly



Disagree) to 7 (Strongly Agree). Items under Usefulness asked whether LaundryLogix helps users accomplish tasks and improves their overall experience; Ease of Use items focused on how simple and effortless it is to operate the system; Ease of Learning items explored how quickly users could become comfortable with the system; and Satisfaction items examined how pleased and confident users felt while using LaundryLogix. After data collection, the mean score for each item and for each dimension was computed, and these means were interpreted using predefined ranges summarized in Table 1, ensuring consistent and transparent interpretation across all results.

Table 1

Mean Interpretation

Weight	Weighted Mean	Interpretation
1	1.00 – 1.85	Strongly Disagree
2	1.86 – 2.71	Disagree
3	2.72 – 3.57	Somewhat Disagree
4	3.58 – 4.42	Neutral
5	4.43 – 5.28	Somewhat Agree
6	5.29 – 6.14	Agree
7	6.15 – 7.00	Strongly Agree

Table 1 presents the scale used to interpret the



computed mean scores from the seven-point Likert-scale responses of the usability questionnaire. Overall, the customers' responses yielded a mean Usefulness score of 5.66, which falls in the "Agree" range, indicating that they generally perceive LaundryLogix as helpful in accomplishing their laundry-related tasks. The highest means reflect strong appreciation for how the system simplifies the process of arranging and monitoring laundry services. The lowest mean suggests that while the system is useful overall, some customers still feel that certain features or details can be further refined to fully match their specific needs.

Table 2

'Usefulness' Customer Respondents

Questions	Mean	Interpretation
It helps me to be more effective.	5.91	Agree
It helps me be more productive.	6.05	Agree
It is useful.	5.68	Agree
It gives me more control over the activities in my life.	5.88	Agree
It makes the things I want to accomplish easier to get done.	6.07	Agree
It saves me time when I use it.	5.77	Agree



It meets my needs.	4.12	Neutral
It does everything I would expect it to do.	5.82	Agree

Summary	5.66	Agree
---------	------	-------

Table 2 details the customers' mean ratings for the individual Usefulness items of the LaundryLogix platform, presenting an overall calculated mean of 5.66 that firmly positions user perception within the "Agree" range and underscores a widespread consensus that the system functions as a genuinely helpful tool in streamlining their laundry-related tasks; within this dataset, the highest mean score of 6.07 was achieved by the statement "It makes the things I want to accomplish easier to get done," highlighting a strong user appreciation for the system's capacity to simplify the coordination and real-time monitoring of laundry services, whereas the lowest mean score of 4.12 was recorded for the phrase "It meets my needs," a metric interpreted as "Neutral" which indicates that while the application is fundamentally effective as a whole, a segment of the customer base believes that specific functionalities or operational features could still be further refined and optimized to perfectly align with their personal requirements.



Table 3

'Ease of Use' Customer Respondents

Questions	Mean	Interpretation
It is easy to use.	6.07	Agree
It is simple to use.	5.99	Agree
It is user friendly.	5.85	Agree
It requires the fewest steps possible to accomplish what I want to do with it.	5.90	Agree
It is flexible.	5.99	Agree
Using it is effortless.	6.02	Agree
I can use it without written instructions.	5.93	Agree
I don't notice inconsistencies as I use it.	6.04	Agree
Both occasional and regular users would like it.	5.90	Agree
I can recover from mistakes quickly and easily.	5.99	Agree
I can use it successfully every time.	6.16	Strongly Agree
Summary	5.99	Agree

Table 3 presents a detailed breakdown of customer



ratings evaluating the system's Ease of Use, which yielded a highly favorable overall mean score of 5.99. This statistical average corresponds directly to the "Agree" designation on the measurement scale, clearly demonstrating that users generally find the LaundryLogix platform to be straightforward, intuitive, and uncomplicated to navigate during their daily operations. Within this data set, the highest level of satisfaction was recorded for the statement *"I can use it successfully every time"* (\$6.16\$), highlighting that customers deeply appreciate the platform's consistent reliability in routine use. Conversely, the lowest score emerged from the statement *"It is user friendly"* (\$5.85\$); while this remains firmly within the positive feedback threshold, it indicates a minor, actionable opportunity for developers to further refine the user interface and make it even more naturally intuitive for all user segments.

Table 4

'Ease of Learning' Customer Respondents

Questions	Mean	Interpretation
I learned to use it quickly.	6.02	Agree
I easily remember how to use it.	5.99	Agree
It is easy to learn to use it.	5.88	Agree



I quickly became skillful with it.	5.79	Agree
------------------------------------	------	-------

Summary	5.92	Agree
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Table 4 displays the customers' ratings for the Ease of Learning dimension, producing an overall mean of 5.92, categorized as "Agree," indicating that most users find LaundryLogix easy to learn even on first use. The highest mean, 6.02, came from the statement "I learned to use it quickly," confirming that customers adapt to the system with minimal training or instruction. The lowest mean, 5.79, was associated with "I quickly became skillful with it," which, although still favorable, suggests that certain functions may benefit from additional cues or guidance to further speed up the learning process for new users.

Table 5

'Satisfaction' Customer Respondents

Questions	Mean	Interpretation
I am satisfied with it.	6.04	Agree
I would recommend it to a friend.	6.24	Strongly Agree
It is fun to use.	5.85	Agree
It works the way I want it to work.	5.88	Agree
It is wonderful.	5.82	Agree
I feel I need to have it.	5.76	Agree



It is pleasant to use.

5.96 Agree

Summary

5.94 Agree

Table 5 presents a comprehensive breakdown of the customer satisfaction metrics, revealing a highly favorable overall composite mean score of 5.94. Within the framework of the evaluation scale, this score aligns precisely with the "Agree" classification, serving as clear empirical evidence that users are generally pleased and content with their user experience when interacting with the LaundryLogix platform. A closer inspection of the data reveals that the highest statistical mean, reaching a notable 6.24, was achieved by the statement *"I would recommend it to a friend,"* which underscores a strong, positive sentiment among the majority of consumers who are highly willing to actively endorse the system to others within their social and professional networks. Conversely, the lowest relative mean within this category was recorded at 5.76 for the statement *"I feel I need to have it."* While this figure still remains securely within the positive feedback threshold, it implies that although customers are highly satisfied with the utility provided, there are still specific, nuanced aspects of the user interface or overall system workflow that can be further optimized to transform



the platform from a helpful utility into an absolutely indispensable tool for their daily operations.

Table 6

'Usefulness' Owner/Staff Respondents

Questions	Mean	Interpretation
It helps me to be more effective.	6.33	Strongly Agree
It helps me be more productive.	6.33	Strongly Agree
It is useful.	6.33	Strongly Agree
It gives me more control over the activities in my life.	6.00	Agree
It makes the things I want to accomplish easier to get done.	5.00	Somewhat Agree
It saves me time when I use it.	6.33	Strongly Agree
It meets my needs.	6.00	Agree
It does everything I would expect it to do.	6.33	Strongly Agree
Summary	6.08	Agree

Table 6 shows the owner and staff respondents' ratings for the Usefulness items of LaundryLogix, with an overall mean of 6.08, which falls in the "Agree" range and indicates that they generally perceive the system as highly helpful in supporting their daily operations. The highest mean, 6.33, was obtained by the statements "It helps me to



be more effective," "It helps me be more productive," "It is useful," "It saves me time when I use it," and "It does everything I would expect it to do," reflecting that internal users strongly value the way LaundryLogix improves effectiveness, productivity, and completeness of features. The lowest mean, 5.00, was recorded for "It makes the things I want to accomplish easier to get done," interpreted as "Somewhat Agree," suggesting that while the system is very useful overall, there is still room to further streamline certain workflows so that every task feels easier to accomplish for the staff and owner.

Table 7

'Ease of Use' Owner/Staff Respondents

Questions	Mean	Interpretation
It is easy to use.	6.33	Strongly Agree
It is simple to use.	6.00	Agree
It is user friendly.	6.33	Strongly Agree
It requires the fewest steps possible to accomplish what I want to do with it.	6.33	Strongly Agree
It is flexible.	6.00	Agree
Using it is effortless.	6.33	Strongly Agree
I can use it without written	6.33	Strongly Agree



instructions.

I don't notice inconsistencies as I use it. 6.00 Agree

Both occasional and regular users would like it. 6.00 Agree

I can recover from mistakes quickly and easily. 6.33 Strongly Agree

I can use it successfully every time. 6.00 Agree

Summary	6.18 Strongly Agree
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Table 7 presents the owner and staff respondents' ratings for the Ease of Use dimension, resulting in an overall mean of 6.18, interpreted as "Strongly Agree," which indicates that they generally find LaundryLogix very straightforward and easy to operate. The highest mean, 6.33, was observed for multiple statements, namely "It is easy to use," "It is user friendly," "It requires the fewest steps possible to accomplish what I want to do with it," "Using it is effortless," "I can use it without written instructions," and "I can recover from mistakes quickly and easily," showing that staff and owner particularly appreciate the simplicity, user-friendliness, and error recovery of the system during routine use. The



lowest mean, 6.00, was associated with the statements "It is simple to use," "It is flexible," "I don't notice inconsistencies as I use it," "Both occasional and regular users would like it," and "I can use it successfully every time," which, while still in the "Agree" range, suggest that a few aspects such as flexibility and perceived consistency can still be refined to make the experience feel even more effortless.

Table 8

'Ease of Learning' Owner/Staff Respondents

Questions	Mean	Interpretation
I learned to use it quickly.	5.33	Agree
I easily remember how to use it.	5.33	Agree
It is easy to learn to use it.	5.33	Agree
I quickly became skillful with it.	6.00	Agree
Summary	5.50	Agree

Table 8 displays the empirical ratings provided by both the business owner and staff respondents regarding the Ease of Learning metric, which yielded a solid overall composite mean score of 5.50. Within the context of the assessment scale, this score falls comfortably into the "Agree" classification, indicating that the system's core target operators were able to understand and navigate the



LaundryLogix platform without experiencing any significant operational difficulty. The highest individual statistical mean reached a notable 6.00 for the statement *"I quickly became skillful with it,"* confirming that once users begin actively engaging with the software, both the owner and floor staff are capable of achieving high proficiency in its various functions within a relatively short period of time. Conversely, the lowest relative means within this category were identical scores of 5.33, recorded for three distinct statements: *"I learned to use it quickly,"* *"I easily remember how to use it,"* and *"It is easy to learn to use it."* While these scores still reside securely within favorable territory, their lower relative standing suggests that implementing a brief initial orientation, dedicated practice sessions, or subtle visual cues within the interface could further accelerate the initial onboarding process and significantly improve long-term user retention of the system's workflows.

Table 9

'Satisfaction' Owner/Staff Respondents

Questions	Mean	Interpretation
I am satisfied with it.	5.33	Agree
I would recommend it to a friend.	7.00	Strongly Agree



It is fun to use.	5.33	Agree
It works the way I want it to work.	5.33	Agree
It is wonderful.	6.00	Agree
I feel I need to have it.	6.00	Agree
It is pleasant to use.	6.33	Strongly Agree
Summary	5.90	Agree

Table 9 details the Satisfaction metrics from the business owner and internal staff, revealing a favorable overall composite mean score of 5.90 that falls within the "Agree" classification, indicating general satisfaction with LaundryLogix as an operational management tool. The data is highlighted by a perfect mean score of 7.00 for the statement *"I would recommend it to a friend"* and a high mean of 6.33 for *"It is pleasant to use,"* underscoring a strong willingness among internal users to endorse the system alongside a highly positive daily user experience. Conversely, the lowest relative scores were identical means of 5.33 for *"I am satisfied with it,"* *"It is fun to use,"* and *"It works the way I want it to work"*. While these values remain securely in favorable territory, they suggest that minor adjustments to user enjoyment and a closer alignment with the staff's ideal, preferred workflows could further elevate their overall satisfaction.



Table 10

'Usefulness' Laundry shops

Questions	Mean	Interpretation
It helps me to be more effective.	6.00	Agree
It helps me be more productive.	5.50	Agree
It is useful.	5.75	Agree
It gives me more control over the activities in my life.	5.75	Agree
It makes the things I want to accomplish easier to get done.	6.00	Agree
It saves me time when I use it.	6.00	Agree
It meets my needs.	5.00	Somewhat Agree
It does everything I would expect it to do.	5.00	Somewhat Agree
Summary	5.63	Agree

Table 10 shows the Usefulness ratings from laundry shop respondents, with a summary mean of 5.63 in the "Agree" range, indicating that they generally consider LaundryLogix useful for their operations. The highest means, 6.00, were obtained by the statements "It helps me to be more effective," "It makes the things I want to accomplish easier to get done," and "It saves me time when I use it," showing that owners and staff from other laundry



shops clearly perceive that LaundryLogix improves efficiency and reduces time spent on routine tasks. The lowest means, 5.00, for "It meets my needs" and "It does everything I would expect it to do" are interpreted as "Somewhat Agree," suggesting that while LaundryLogix is useful overall, there is still room for further enhancement so that it fully satisfies the detailed needs and expectations of other laundry businesses.

Table 11

'Ease of Use' Laundry shops

Questions	Mean	Interpretation
It is easy to use.	5.50	Agree
It is simple to use.	6.00	Agree
It is user friendly.	5.75	Agree
It requires the fewest steps possible to accomplish what I want to do with it.	5.50	Agree
It is flexible.	5.25	Somewhat Agree
Using it is effortless.	6.00	Agree
I can use it without written instructions.	5.75	Agree
I don't notice inconsistencies as I use it.	6.25	Strongly Agree



Both occasional and regular users	5.75	Agree
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would like it.

I can recover from mistakes quickly	5.50	Agree
-------------------------------------	------	-------

and easily.

I can use it successfully every	6.00	Agree
---------------------------------	------	-------

time.

Summary	5.75	Agree
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Table 11 presents the Ease of Use ratings from the laundry shop respondents, yielding a solid summary mean score of 5.75 that falls within the "Agree" classification and indicates they generally find LaundryLogix straightforward and manageable to operate. The data is highlighted by a peak mean score of 6.25 for the statement *"I don't notice inconsistencies as I use it,"* suggesting that users experience a highly consistent, predictable interface and workflow when interacting with the system. Conversely, the lowest relative mean score of 5.25 was recorded for *"It is flexible,"* which places it in the "Somewhat Agree" range. While this still represents a favorable assessment, it implies that although the platform is easy to navigate, certain laundry shop users would appreciate greater flexibility to customize and adapt the system to their specific operational processes or



individual preferences.

Table 12

'Ease of Learning' Laundry shops

Questions	Mean	Interpretation
I learned to use it quickly.	6.00	Agree
I easily remember how to use it.	5.75	Agree
It is easy to learn to use it.	6.00	Agree
I quickly became skillful with it.	5.75	Agree
Summary	5.88	Agree

Table 12 displays the Ease of Learning ratings from the laundry shop respondents, resulting in a favorable summary mean score of 5.88 that is interpreted as "Agree," indicating they generally find LaundryLogix easy and accessible to learn. The highest individual mean score of 6.00 was achieved by the statements "*I learned to use it quickly*" and "*It is easy to learn to use it,*" confirming that laundry shop personnel can successfully become familiar with the system's core functions in a short amount of time. Conversely, the lowest relative means within this domain were identical scores of 5.75 for the statements "*I easily remember how to use it*" and "*I quickly became skillful with it*". While these figures still remain securely within the positive "Agree" range, they suggest



that while the platform is simple to pick up initially, continued everyday exposure and regular practice will further strengthen users' operational confidence and overall skill retention.

Table 13

'Satisfaction' Laundry shops

Questions	Mean	Interpretation
I am satisfied with it.	5.50	Agree
I would recommend it to a friend.	5.75	Agree
It is fun to use.	5.75	Agree
It works the way I want it to work.	5.75	Agree
It is wonderful.	6.00	Agree
I feel I need to have it.	5.50	Agree
It is pleasant to use.	6.00	Agree
Summary	5.75	Agree

Table 13 shows the Satisfaction ratings of the laundry shop respondents, yielding a solid summary mean score of 5.75 that corresponds to "Agree" and indicates they are generally satisfied with LaundryLogix as a system they could implement in their own businesses. The highest individual mean scores of 6.00 were recorded for the statements "*It is wonderful*" and "*It is pleasant to use,*" showing that respondents particularly appreciate the



overall user experience and feel highly positive when interacting with the system. Conversely, the lowest relative means within this category were identical scores of 5.50 for "*I am satisfied with it*" and "*I feel I need to have it.*" While these values remain securely within the favorable "Agree" range, they suggest that while baseline satisfaction is high, there is a distinct potential to further strengthen the perceived necessity of the software by continuing to refine features and tailoring them more closely to the unique operational contexts of various laundry shops.



Chapter 4

CONCLUSIONS AND RECOMMENDATIONS

This chapter presents the conclusions and recommendations of the study based on the information requirements identified, the architectural framework designed, the features implemented in the LaundryLogix system, and the results of the usability evaluation conducted among the customers, staff, and owner of Elyssa Laundry Center.

Conclusions

This section provides the conclusions based on the findings obtained from the study.

1. The researchers conclude that clearly defining information requirements for users, transactions, machines, orders, notifications, and inventory is essential for LaundryLogix to support efficient operations and accurately reflect Elyssa Laundry Center's workflow.

2. The researchers conclude that a centralized, cloud-based architecture guided by the 4+1 Architectural View Model and UML diagrams provides a coherent structure aligned with the shop's functional, behavioral, development, and physical needs.

3. The researchers conclude that LaundryLogix



features for POS machine management, stock monitoring, pricing, transaction history, analytics, pickup and delivery, reservations, digital payments, and the customer mobile app effectively convert manual processes into integrated digital workflows that improve transparency, organization, and convenience.

4. The study found that LaundryLogix received very favorable USE ratings from customers, owner/staff, and laundry shop respondents, with questions such as "It is easy to use," "It is user friendly," "Using it is effortless," and "I would recommend it to a friend" obtaining the highest mean scores in the scale, indicating that the system is useful, easy to use, easy to learn, and satisfying.

Recommendations

In light of the findings and conclusions of the study, the following recommendations are presented for future researchers and system developers.

1. It is recommended that future researchers and developers further analyze information requirements, especially user, transactional, and resource data, to refine LaundryLogix or similar systems.

2. It is recommended that future versions enhance



the interface by organizing the mobile app into clearer tabs (Services and Prices, Machines, Self-Service vs. Full-Service), highlighting extras and reservation prices, and adding options like a second address field and automatic status labels for customer requests.

3. It is recommended that future iterations improve user experience by refining color coding, standardizing fonts and sizes, enlarging critical text where needed, considering a dark mode, and using less intrusive bottom notifications instead of center pop-ups.

4. It is recommended to the response of the lower-rated items in the USE Questionnaire, the researchers recommend enhancing system flexibility, simplifying workflows, and improving the functions that users felt did not fully meet their needs, especially in task completion and configuration options.



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APPENDICES

Appendix A

Communication Letter



November 24, 2025

Ms. Elayssa F. Salgado
Business Owner
Elyssa Laundry Center
Legarda Road, Baguio City

Dear Ms. Salgado:

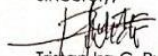
In partial fulfillment of the requirements for the degree Bachelor of Science in Information Technology, graduating students are required to undertake a study that involves the development of an information system. This undertaking is intended to expose students to actual practices in systems development.

In line with the above, the undersigned wish to request approval from your office to study your Point-of-Sale System. Should this request be approved, data-gathering activities will be conducted at a time deemed most convenient for you. All data gathered will be used strictly for academic purposes only and will be treated with utmost confidentiality.

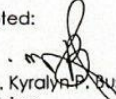
Further, an acceptance test shall be conducted to validate that the developed system conforms to the given requirements.

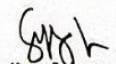
Look forward to a positive response.

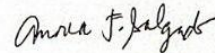
Sincerely,


Tristan Ira C. Pulido
Project Leader

Noted:


Ms. Kyrallyn P. Busaing
Adviser


Dr. Jeffrey S. Ingosaran
Academic Dean, CICTS





Appendix B

Certificate of Acceptance





Appendix C

Interview Questions

1. When a machine suddenly breaks down in the middle of a wash or dry cycle, how do you usually handle refunds, voiding the transaction, or making up for it to the customer?
 2. Do you have any kind of loyalty program for suki customers (like discounts, promo pricing, or stamp cards)? If yes, how do you record or track those right now?
 3. For full-service laundry, do customers normally pay the full amount when they drop off, or do you also accept partial down payments?
 4. How do you usually measure and sell your consumables—for example, is it per sachet, per scoop, or by pump/milliliter from a bigger container?
 5. For each supply (detergent, fabric conditioner, plastic bags, etc.), at what remaining level do you want the system to warn you that it's time to reorder?
 6. How do you currently monitor the cash drawer during the shift and when changing shifts, and what records do you keep to reconcile the cash at the end of the day?
 7. Which actions in the system do you want to be owner-only, like changing prices, editing completed transactions, or viewing detailed reports?
 8. What is your current policy for unclaimed or overdue laundry, and should the system automatically tag orders that haven't been picked up after a certain number of days?
 9. Would you like the system to automatically send notifications to customers whenever their laundry status changes (for example, washing started, drying finished, folded, ready for pickup)?
 10. For your end-of-day summary, what specific details do you want to see (like total sales, number of transactions, cash vs digital payments, refunds, or discounts)?
 11. Are there services or add-ons (such as express service, stain treatment, or special handling for certain fabrics) that should have different pricing rules or cut-off times in the system?
 12. How do you currently handle customer complaints about missing or damaged items, and would it be helpful if the system could record notes or incident logs for these cases?
 13. Do you notice any seasonal patterns in your business (like enrollment season, holidays, or rainy months) that you want the system's reports or analytics to highlight for planning purposes?
-



Appendix E

Sample of Answered Usability Questionnaire

USEFULNESS	EASE OF USE
This section measures how helpful and beneficial the LaundryLogix customer mobile application is for managing your laundry transactions. It focuses on whether the app helps you track your laundry more easily, stay informed about its status, save time, and accomplish what you want to do as a customer.	This section evaluates how simple and convenient the LaundryLogix mobile application is to operate. It looks at whether the app is easy to navigate, user-friendly on your phone, requires only a few steps to complete tasks, feels effortless to use, and allows you to recover from mistakes or wrong taps without difficulty.
It helps me to be more effective. 1 2 3 4 5 6 7 Strongly Disagree ☐ ☐ ☐ ☐ ☐ ☒ ☐ Strongly Agree	It is easy to use. 1 2 3 4 5 6 7 Strongly Disagree ☐ ☐ ☐ ☐ ☐ ☒ ☐ Strongly Agree
It helps me be more productive. 1 2 3 4 5 6 7 Strongly Disagree ☐ ☐ ☐ ☐ ☐ ☐ ☒ Strongly Agree	It is simple to use. 1 2 3 4 5 6 7 Strongly Disagree ☐ ☐ ☐ ☐ ☐ ☐ ☒ Strongly Agree
It is useful. 1 2 3 4 5 6 7 Strongly Disagree ☐ ☐ ☐ ☐ ☒ ☐ ☐ Strongly Agree	It is user friendly. 1 2 3 4 5 6 7 Strongly Disagree ☐ ☐ ☐ ☐ ☒ ☐ ☐ Strongly Agree
EASE OF LEARNING This section assesses how quickly and easily you learned to use the LaundryLogix mobile application. It focuses on how fast you understood the screens and buttons, how easily you remember how to use the app after some time, and how quickly you became comfortable using its features.	SATISFACTION This section measures your overall satisfaction and feelings about using the LaundryLogix mobile application. It looks at whether you are satisfied with the app, would recommend it to others, find it pleasant or enjoyable to use, and feel that it works the way you want when managing your laundry with Elyssa Laundry Center.
I learned to use it quickly. 1 2 3 4 5 6 7 Strongly Disagree ☐ ☐ ☐ ☐ ☒ ☐ ☐ Strongly Agree	I am satisfied with it. 1 2 3 4 5 6 7 Strongly Disagree ☐ ☐ ☐ ☒ ☐ ☐ ☐ Strongly Agree
I easily remember how to use it. 1 2 3 4 5 6 7 Strongly Disagree ☐ ☐ ☐ ☐ ☐ ☒ ☐ Strongly Agree	I would recommend it to a friend. 1 2 3 4 5 6 7 Strongly Disagree ☐ ☐ ☐ ☐ ☐ ☒ ☐ Strongly Agree
It is easy to learn to use it. 1 2 3 4 5 6 7 Strongly Disagree ☐ ☐ ☐ ☐ ☐ ☐ ☒ Strongly Agree	It is fun to use. 1 2 3 4 5 6 7 Strongly Disagree ☐ ☐ ☐ ☐ ☐ ☒ ☐ Strongly Agree



Appendix F

Documentation





Appendix G

LaundryLogix User Manual

ELYSSA LAUNDRY CENTER | Quick Start Guide

1 GETTING STARTED (Login & Access)



- Mockup mockup screen | the staff login page (like phone or POS screen)
 - Enter the vain details and s licks the staff page
 - Password your names or requeston on manual
- 🔒 Set to Staff starts en-to-pos instruction or bont, a Secure daddock illustration or resourts.

2 MACHINE DASHBOARD



3 PROCESSING A LOAD

3a) Enter Customer Details

Name	
Phone	
Weight (kg)	
Start	

3b) Select Service Type

☒ Self Service Self Service or Full Service
☐ Full Service Self Services or Full Service

3c) Add Consumables

Detergent	P20
Fabric conditioner	\$20
Price	P150

4 MANAGING MOBILE REQUESTS



- Staff hand selects a request to process.
- Staff hand selects a request to process.

5 DAILY LOGS & PAYMENT

5a) Receive Payment



5b) Daily Logs

Summary Table	
Total Transactions	11
Total Income	Php 2,850
Daily Gross:	Php 2,850



Appendix H

POS System Source Code

```
page.tsx X
laundrylogixweb > src > app > setup-machines > page.tsx > ...

1
2 'use client';
3
4 import { useState } from 'react';
5 import { useFirebase } from '@/firebase/provider';
6 import { writeBatch, doc } from 'firebase/firestore';
7 import { Button } from '@components/ui/button';
8 import { Card, CardContent, CardDescription, CardHeader, CardTitle } from '@components/ui/card';
9 import { Alert, AlertDescription, AlertTitle } from '@components/ui/alert';
10 import { CheckCircle, AlertTriangle } from 'lucide-react';
11 import { useRouter } from 'next/navigation';
12
13 const initialWashers = Array.from({ length: 7 }, (_, i) => ({
14   id: `washer-${i + 1}`,
15   number: i + 1,
16   type: 'washer',
17   status: 'Available',
18 }));
19
20 const initialDryers = Array.from({ length: 7 }, (_, i) => ({
21   id: `dryer-${i + 1}`,
22   number: i + 1,
23   type: 'dryer',
24   status: 'Available',
25 }));
26
27 const allMachines = [...initialWashers, ...initialDryers];
28
29 export default function SetupMachinesPage() {
30   const { firestore } = useFirebase();
31   const router = useRouter();
32   const [isLoading, setIsLoading] = useState(false);
33   const [error, setError] = useState<string | null>(null);
34   const [success, setSuccess] = useState(false);
```

Appendix I



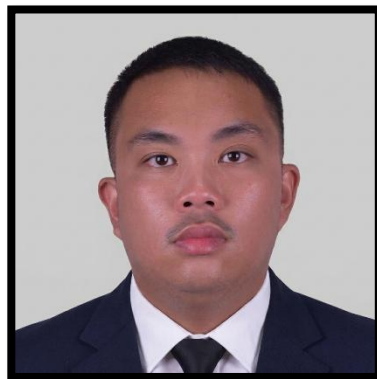
CURRICULUM VITAE

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EDUCATIONAL ATTAINMENT

Tertiary	: Bachelor of Science in Information Technology : Network & Security Track : University of the Cordilleras : Gov. Pack Rd., Baguio City : Expected Date of Graduation: October 2026
Secondary	: Northridge Academy Inc. : 001 Camella St. Upper QM : March 2012
Primary	: Northridge Academy Inc. : 001 Camella St. Upper QM : March 2008

SKILLS AND QUALIFICATIONS

- Programming Languages: Python, JavaScript, Java, C++, HTML/CSS, SQL
 - Tools & Technologies: MySQL, MongoDB, AWS, Azure, Github
 - Networking & OS: Networking protocols, Windows, Linux
 - Cybersecurity & Testing: Basic knowledge of cybersecurity and software testing
-



- Soft Skills: Problem-solving, communication, teamwork, time management
- Knowledgeable in CISCO LAN Switches and routers installation and configuration (CCNA 1 to CCNA 4) using PuTTY and Packet Tracer
- Experienced in hardware prototyping and development using Arduino
- Motivated with a strong willingness to learn and adapt to new technologies

SEMINARS and TRAININGS ATTENDED

- CITCS Webinar Series
Practical App Development
March 14, 2026
Zoom Meeting
- CITCS Webinar Series
Practical Data Analytics Using Python and Data Visualization
March 14, 2026
Zoom Meeting
- DevFest Baguio 2025
October 18, 2025
Baguio Convention and Cultural Center, Baguio City Philippines

PERSONAL INFORMATION

Date of Birth	: October 29, 1995
Age	: 30 years old
Height	: 5'5"
Weight	: 90 kgs.



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EDUCATIONAL ATTAINMENT

Tertiary	: Bachelor of Science in Information Technology : Network & Security Track : University of the Cordilleras : Gov. Pack Rd., Baguio City : Expected Date of Graduation: October 2026
Secondary	: Baguio City High School : Gov. Pack Rd., Baguio City : April 2017
Primary	: Baguio Central School : Yandoc Street, Baguio City : March 2013

SKILLS AND QUALIFICATIONS

- Proficient in programming languages including Java, Python, and C++
- Knowledgeable in CISCO LAN Switches and routers installation and configuration (CCNA 1 to CCNA 4) using PuTTY and Packet Tracer
- Experienced in hardware prototyping and development using Arduino
- Advanced user of office and creative software: MS Word, Excel,



PowerPoint, Canva, and DaVinci Resolve

- Highly motivated with a strong willingness to learn and adapt to new technologies
- Strong verbal and personal communication skills

SEMINARS and TRAININGS ATTENDED

- DevFest Baguio 2025
October 18, 2025
Baguio Convention and Cultural Center
- Webinar Series: Practical Data Analytics Using Python and Data Visualization
March 14, 2026
University of the Cordilleras
- Webinar Series: Practical App Development
March 14, 2026
University of the Cordilleras

PERSONAL INFORMATION

Date of Birth	: July 29, 2000
Age	: 25 years old
Height	: 5'5"
Weight	: 80 kgs.



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WORK EXPERIENCE

Labor(Machine Operator)	: Purebreed Poultry Co. AlJithamiyah, Ha'il, Kingdom of Saudi Arabia : January 2021, April 2023
Geologic Aide 1	:Eds Construction Philes Mines Padcal, Tuba Benguet : January 2017, December 2017
Core Checker	: Philex Mining Corporation : January 2016, December 2016
Geologic Helper	: Philex Mining Corporation : March 2015, May 2015
Utility	: Philex Mining Corporation : May 14, October 2014
Geologic Aide	: Aritao Contracts : October 2013, January 2014
Badminton Trainer	: Alaminos City (Local Government Unit) : August 2009, August 2013

EDUCATIONAL ATTAINMENT

Tertiary	: Bachelor of Science in Information Technology : Network & Security Track : University of the Cordilleras : Gov. Pack Rd., Baguio City
-----------------	--



	: Expected Date of Graduation: October 2026
Secondary	: Saint Louis High School Philex
	: Philex Mines Padcal, Tuba, Benguet
	: April 2003
Primary	: Philex Mines Elementary School
	: Philex Mines Padcal, Tuba, Benguet
	: March 1999

SKILLS AND QUALIFICATIONS

- Programming : C++, Java, Python, HTML, CSS, Tailwind CSS, Node.js, JavaScript, React JS, JavaScript, Tinkercad, MongoDB, NoSQL. Knowledgeable in CISCO LAN Switches and routers, installation and configuration (CCNA1 to CCNA 4)
- Networking : Wireless, Routers, Switches, PuTTY (SSH) and Wireshark (include IT certifications if any, e.g. CISCO Certified Network Associate)
- Operating Systems : Linux, Windows (Windows)
- Control : Arduino
- Office Tools : Excel (VLOOKUP), PowerPoint, Canva, (AutocadCAD)
- Language : English, Tagalog and Ilocano

SEMINARS and TRAININGS ATTENDED

- Build a Full Website using WordPress
March 1, 2024
- Get Started with Figma
Feb 16, 2024
- CCNAV7: Introduction to Networks Cisco Networking Academy –
Issued Apr 2024
20 Apr 2024
- Autocad 2d
GMS Technology Baguio City
January 15, 2018 - February 8, 2018



- Autocad 3d
GMS Technology Baguio City
february 12 ,2018 - March 7, 2018
- Webinar Series: Practical Data Analytics Using Python and Data
Visualization
March 14, 2026
University of the Cordilleras
- Webinar Series: Practical App Development
March 14, 2026
University of the Cordilleras

PERSONAL INFORMATION

Date of Birth	: September 29, 1986
Age	: 39 years old
Height	: 5'4"
Weight	: 56 kgs.